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Retrieval of land surface properties from an annual time series of Landsat TOA radiances during a drought episode using coupled radiative transfer models

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ABSTRACT

The accurate retrieval of land surface vegetation properties under varying environmental conditions from time series of moderately high spatial resolution satellite observations is challenging. By coupling various Radiative Transfer (RT) models one can describe the soil, vegetation and atmosphere contributions in a "bottom-up" approach and, thereby, simulate top-of-atmosphere (TOA) spectral radiance data comparable to satellite-observed TOA radiances. This makes it possible to retrieve vegetation properties directly from TOA radiances rather than from atmospherically corrected top-of-canopy (TOC) reflectance data. The advantages of this approach are that a separate atmospheric correction of the satellite images is not necessary, and that the anisotropic surface reflection can also be taken into account effectively. In this study, we coupled various RT models, including the brightness – shape – moisture (BSM) reflectance model of the soil, the optical radiative transfer (RTMo) model of vegetation and the 'MODerate resolution atmospheric TRANsmission' (MODTRAN) model of the atmosphere, to simulate an annual time series of Landsat satellite TOA radiances observed during a drought episode in California Mediterranean grasslands in 2004. The inversion of this coupled system through an optimization technique against Landsat TOA radiances resulted in direct retrieval of vegetation properties. We accommodated the surface anisotropic reflection in our coupled modeling and also defined a novel anisotropy index to quantitatively express the importance of this phenomenon in satellite image analysis for the first time. Our study showed that the coupled use of RT models was able to accurately reproduce the time series of observed TOA radiances collected under varying soil moisture contents during an extended drought episode. The proposed coupling approach is useful for successful retrieval of vegetation properties from time series of satellite TOA radiance data to produce maps of land surface properties and to monitor vegetation properties variations in an operational straightforward way. The approach can also be easily adapted for conducting multi-sensor time series studies, creating a much denser temporal sampling than would be possible for separate single sensors.

1. Introduction

Satellite time series observations, when combined in a quantitative modeling approach, can provide a unique opportunity for operational monitoring of land surface (i.e., vegetation) properties under various environmental and atmospheric conditions. There are mainly two quantitative data processing approaches which have been employed for translating satellite observations into vegetation properties information (as shown in Fig. 1). The first is to use vegetation indices (VIs) in a statistical approach (Ceccato, 2002; Chávez et al., 2013; Darvishzadeh et al., 2011; Houborg et al., 2007; Zarco-Tejada et al., 2013) and the second is to use a vegetation radiative transfer (RT) model in a physical modeling approach (Bayat et al., 2016a, 2016b; Bicheron and Leroy, 1999; Clevers et al., 2010; Darvishzadeh et al., 2008; Dorigo et al., 2009; Duan et al., 2014; Houborg et al., 2007; Jacquemoud et al., 1995; Laurent et al., 2011a; Schaepman-Strub et al., 2006; Verhoef, 1985, 1984; Verhoef and Bach, 2007, 2003a; Zhang et al., 2014; Zhang et al., 2013; Zhang et al., 2012; Zhang et al., 2009; Zhang et al., 2005). VIs have the advantage that they can be applied very easily and that they can have a reasonable correlation with vegetation properties, such as leaf area index (*LAI*) and leaf chlorophyll content (C_{ab}) (Darvishzadeh et al., 2011). However, VIs may be affected by several factors simultaneously, which implies that the explored relationships between VIs and surface properties are species and observation specific and thus cannot be applied universally. In addition, computation of VIs is based on employing only a few of the available bands. As a consequence,

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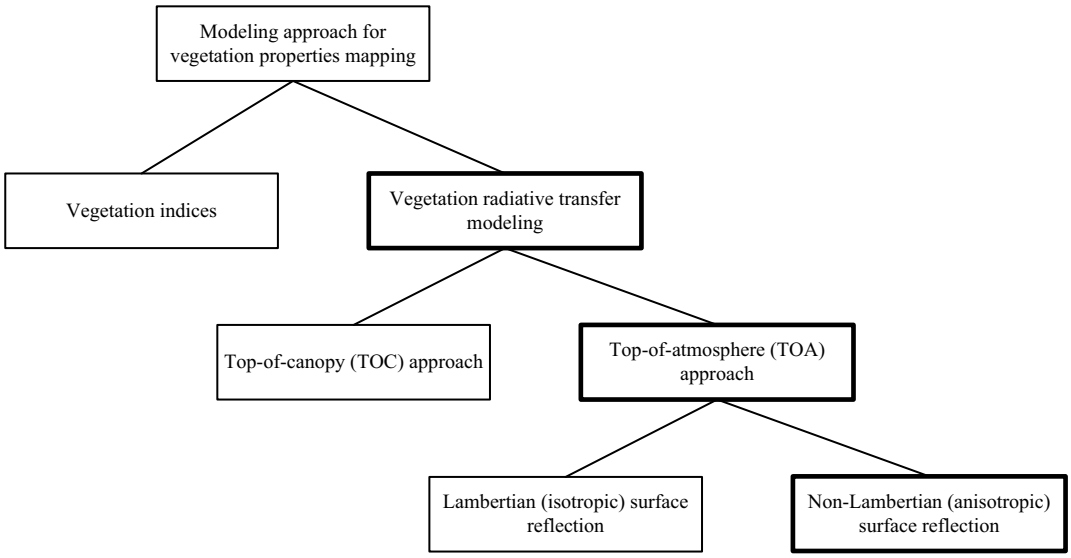


Fig. 1. Quantitative modeling approach for mapping spatio-temporal variations of vegetation properties. The bolded outline boxes represent the approach adapted in this study.

valuable pieces of information embedded in the other bands are ignored. The alternative, that of using physical vegetation RT models, has these advantages: that; (1) they offer explicit relationships between the whole spectrum and vegetation properties (Atzberger et al., 2011; Clevers et al., 2010), (2) they often yield a more accurate estimation of vegetation properties compared to VIs (Le Maire et al., 2011), and (3) produce more generalizable outputs while being less dependent on in situ measurements (Atzberger et al., 2011; Baret and Buis, 2008). Thus, using a canopy RT model, albeit more complicated, is considered a better way to explore satellite observations efficiently and to retrieve vegetation properties more accurately.

Most vegetation RT models produce surface or top of canopy (TOC) reflectance, and are therefore suitable to retrieve vegetation properties from TOC observations (the 'TOC approach'). This approach has been applied in many studies successfully, resulting in the retrieval of vegetation properties with acceptable accuracy. However, satellite observations are originally recorded at the top of the atmosphere (TOA), so that still an atmospheric correction is required to obtain the TOC reflectance from the observed TOA radiances. Applying a reliable atmospheric correction to satellite observations to obtain the TOC reflectance, however, is still a challenging and cumbersome task (Laurent et al., 2011b), posing considerable limitations especially when the analysis of time series of images is desired. Further, the retrieval of vegetation properties might be subject to errors not only due to modeling uncertainty but also to errors in the computed TOC reflectance caused by inaccurate atmospheric correction assumptions. On the other hand, Level-2 surface reflectance products, which are already corrected atmospherically, are not commonly (or at least not freely) available for new sensors, and the fact remains that the quality of such products is still questionable since they are based on the assumption that both target and surroundings are Lambertian (isotropic) reflectors, which is inconsistent with what is known about the anisotropy in the reflectances of vegetation, bare soil and water bodies.

Since considerable limitations of the TOC approach are related to performing the atmospheric correction to convert TOA radiance to TOC reflectance, this appears to be unsuitable as a feasible and operational set-up for time series analysis. An alternative is to use the TOA radiance data directly for the retrieval (Verhoef et al., 2018; Verhoef and Bach, 2003a). This requires simulation of the surface - atmosphere system by coupling a set of physical RT models describing various objects from soil, to leaf, canopy, and to the atmosphere. Such a coupling approach has the crucial advantage that one can also take the anisotropic surface reflection into account in the forward RT modeling. This is important due to the fact that the majority of vegetation canopies are clearly non-Lambertian (Verhoef et al., 2018). Therefore, the forward modeling of this coupled RT system will produce more accurate TOA radiance results. Further, since the coupled system produces the TOA radiance, which is indeed the same physical quantity as observed by the satellite sensors, justice is done to the actual physics of the observations. By including specific information about particular satellite sensors (e.g., spectral bands, spectral response function, spectral resolution) in the modeling, the simulated TOA radiance data are directly comparable to the ones observed by the satellite sensor (Verhoef and Bach, 2003b). Thus, by the coupled use of an atmospheric RT with a vegetation RT and a soil reflectance model, one may simulate TOA radiance and retrieve land surface properties directly by inversion of such a coupled model (Verhoef et al., 2018).

Several efforts have been made towards accomplishing such a coupling approach to simulate TOA radiance images for different sensors (Börner et al., 2001; Faurtyot and Baret, 1997; Gastellu-Etchegorry et al., 2004; Laurent et al., 2011b; Laurent et al., 2011a; Rahman et al., 1993; Verhoef et al., 2018; Verhoef and Bach, 2012, 2007, 2003b). Among them, (Verhoef and Bach, 2003b) coupled a leaf model (PROSPECT), a canopy two-layer model (GeoSAIL) and the MODerate resolution atmospheric TRANsmission (MODTRAN) model to simulate hyperspectral and directional radiance images over vegetated areas and bare fields. Similarly, (Verhoef and Bach, 2007) coupled the soil-leaf-canopy (SLC) model to MODTRAN4 in order to simulate the hyperspectral multi-angular surface reflectance and TOA radiance for the multi-angular CHRIS sensor. They compared the simulated surface reflectances with the ones observed with the CHRIS sensor for soil, maize, dense and less dense forest pixels, and concluded that spectral-directional observations of CHRIS could be reproduced by the SLC model with good accuracy. Furthermore, (Verhoef and Bach, 2012) coupled the models SLC (for modeling soil and vegetated pixels), WASI (for modeling inland water pixels) and MODTRAN4 (for modeling atmospheric effects) to simulate TOA radiance spectra of Sentinel-3 observations. A similar combination of SLC and MODTRAN was inverted against CHRIS multi-angular TOA radiances on three Norway spruce stands (Czech Republic) to retrieve forest variables in 2006 (Laurent et al., 2011a). Most recently, (Verhoef et al., 2018) used a novel set of coupled RT models of the soil (brightness - shape - moisture (BSM) model), canopy (the Soil-Canopy Observation Photosynthesis and Energy fluxes (SCOPE) model) and atmosphere (MODTRAN model) to

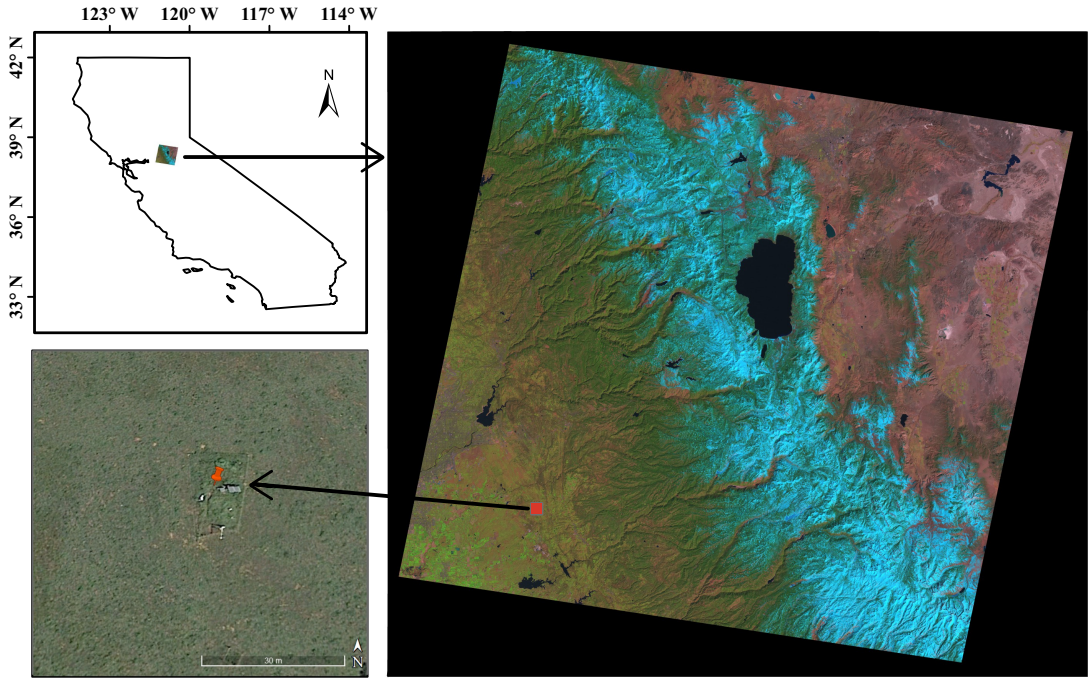


Fig. 2. Study site (Vaira Ranch, in California) representing the footprint of Landsat images (WRS-2 path/row: 043/033). The Landsat image acquired on 15 March 2004 (color composite of red = band 5, green = band 4 and blue = band 3) is shown in the right panel in which the Vaira site location is indicated by a red square. The Vaira site and its surroundings, exported from the Google Earth images, are also shown in the bottom left panel. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

simulate TOA radiance data of several optical sensors of the Fluorescence Explorer (FLEX)/Sentinel-3 (S3) tandem mission. They concluded that one can retrieve surface fluorescence and vegetation biophysical properties simultaneously and with remarkable accuracy from TOA radiance data.

The main focus of the described studies was to chain different RT models to simulate the TOA directional radiance of an image at a specific time (as a snapshot) and, therefore, (1) the background soil reflectance contribution was considered relatively stable, and did not vary due to the fact that soil properties (e.g., soil moisture content) were fixed, (2) the input parameters of the vegetation RT models were obtained either from a land cover map plus ground measurements at the desired time (day), or by running RT models in forward mode and making a database of reflectance, (3) the atmospheric properties (e.g., visibility, water vapor and aerosol models) did not vary much because of the relatively stable atmospheric conditions over a short period (one day), and (4) the radiance of an airborne hyperspectral image or future planned satellite mission could be simulated. Therefore, in the majority of the previous studies, the coupled models were applied to simulate a specific satellite or airborne hyperspectral image for a single flight path. However, the potential of such a coupled model for wider and operational applications is still unexplored and has not been dealt with in depth. Monitoring the effects of soil moisture deficit on vegetation properties during a prolonged drought episode, for example, requires time series of moderately high spatial resolution observations over a relatively long period. In this situation, not only environmental (soil and vegetation properties) but also atmospheric conditions [visibility (Vis) and aerosol models (Aer)] may vary day by day, which consequently stresses the need to consider soil, vegetation, and atmospheric variations, sufficiently. In other words, there is still a strong interest for exploring this TOA coupling approach for analyzing time series of available satellite observations, acquired during a longer episode in which the soil, vegetation, and atmospheric properties are all changing. The current study is focused particularly on this aspect.

We examine the ability of coupled RT models to capture changes in the soil-vegetation-atmosphere system over time to produce accurate and consistent results during a longer episode. Furthermore, we intend to investigate the importance of taking surface anisotropic reflection into account since more recent evidence (Verhoef et al., 2018) demonstrates the importance of non-Lambertian surface reflection. In this context, not only have we considered surface anisotropic reflection in our coupled model, but we also defined a novel anisotropy index to express the importance of this phenomenon. We incorporate this anisotropy effect in time series image analysis quantitatively applied to satellite data for the first time. The combination of RT models we used to describe the soil – vegetation – atmosphere system in a bottom-up approach, comprises the BSM soil reflectance model, the vegetation optical radiative transfer (RTMo) model of the SCOPE and the MODTRAN version 5.2.1 atmospheric RT model. The canopy RT model produces four anisotropic surface reflectance factors, and the sensor's spectral sampling is incorporated in the forward RT modeling as well. The best match obtained between simulated TOA radiances and satellite-observed TOA radiances resulted in the direct retrieval of vegetation properties [notably *LAI*, C_{ab}, leaf water content (C_w), leaf dry matter content (C_{dm}), the senescent material content (C_s) and the leaf inclination distribution function (*LIDF*)] by model inversion through optimization. We applied this approach to a time series of twenty-four Landsat (TM5 and ETM7) observations acquired during a prolonged drought episode in California Mediterranean grasslands in 2004 to map spatio-temporal variations of vegetation properties and explore the trend of retrieved properties during this episode.

2. Data

2.1. Site description

The study focused on the Vaira Ranch (US-Var) Fluxnet site (Fig. 2), hereafter called Vaira site, in the United States of America (Baldocchi et al., 2004; Ma et al., 2011; Xu and Baldocchi, 2004), which is an open grassland, in a Mediterranean climate at 129 m altitude near the foothills of the Sierra Nevada mountain range in California (38.4133° N; 120.9508° W).

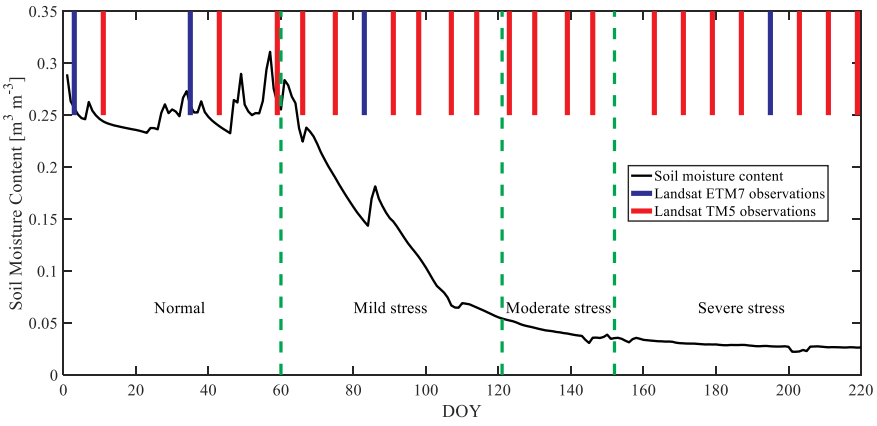


Fig. 3. Timing of satellite overpass dates during the soil moisture deficit episode. Landsat TM5 (thick red lines from top) and Landsat ETM7 (thick blue lines from top) observations track covering a soil moisture deficit (black curve) episode at the Vaira site. The episodes are divided into four periods (separated by dotted green lines) indicating normal, mild stress, moderate stress, and severe stress conditions. These periods within the time series are based on the Palmer Drought Severity Index (PDSI) data set published by NOAA's National Centers for Environmental Information (NCEI). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The mean annual temperature and precipitation in the region are 16.6 °C and 559 mm, respectively. A prolonged drought occurred at the site in 2004, and it was reported as one of the driest years there (Ma et al., 2007). The soil moisture content declined 10-fold from 0.3 to 0.03 $m^3 m^{-3}$ during a period of 220 days (Fig. 3). Further, The NOAA's National Centers for Environmental Information (NCEI) reported variations from 0.44 to −3.25 for the Palmer Drought Severity Index (PDSI) at the Vaira site in 2004 (www.ncdc.noaa.gov/temp-and-precip/drought/historical-palmers). PDSI values suggest near-normal conditions at the site from the first of January (DOY 1) until the end of February (DOY 60), while drought levels with increasingly higher severity, i.e., mild, moderate and severe, were recorded from the first of March (DOY 61) till the end of August (DOY 220), as shown in Fig. 3.

2.2. Vegetation characteristics at the site

The land cover of the Vaira site is cool-season C3 (corresponding to the 3-carbon photosynthetic pathways) grass with the dominant species *Brachypodium distachyon*, *Erodium cicutarium*, *Bromus hordeaceous*, *Hypochaeris glabra*, *Trifolium dubium Sibth*, *Trifolium hirtum*, *Erodium botrys* and *Dichelostemma volubile* (Xu and Baldocchi, 2004). The grass species composition might vary slightly from one year to another based on the climatic conditions (i.e., precipitation and temperature) of the area (Heady, 1958; Jackson, 1985; Pitt and Heady, 1978). The grassland is physiologically functioning from the autumn (e.g., November) to the next spring (e.g., May), with the peak growth period in late February to early April. In this period, the Vaira site contains multiple species belonging to various functional groups (i.e., grasses, forbs and nitrogen fixers) (McNaughton, 1968) which make up a closed canopy (Baldocchi et al., 2004). The average height of the grass during the season was recorded as 0.3 m from 2001 to 2004, while, the maximum grass height may reach 0.55 ± 0.12 m.

2.3. Remote sensing observations

Twenty Landsat TM5 and four Landsat ETM7 observations (WRS-2 path/row: 043/033), hereafter all termed Landsat, were selected such that a soil moisture deficit episode was covered (Fig. 3). The Landsat data used in this study obtained from the US Geological Survey (USGS) website (https://glovis.usgs.gov/). Such data sets were standard L1T products in DN format and collected under nadir view angle at mid-morning equatorial overpass time (between 10:18 to 10:28 a.m.). The quality of the Landsat images used in this study was identified as 'best' (code 9 in image metadata) with completely clear sky over the flux site for all days of the overpass.

2.4. Ground measurements

2.4.1. Soil moisture measurements

Volumetric soil moisture (SM) was measured at the site with a frequency-domain reflectometer probe (ML2x, Delta-T Devices, Burwell, Cambridge, UK) at 2, 10 and 20 cm depths. These sensors were automatically sampled by data loggers (CR10X or CR23, Campbell Scientific, Inc., Logan, Utah, USA) every 10 s, and their half-hour averages were recorded. In this study, we used time series of soil moisture content measured at 2 cm depth.

2.4.2. LAI measurements

LAI was measured destructively with intervals of 2–3 weeks during this drought episode. Grass was collected from three sample plots (0.2 m × 0.2 m) in the prevailing direction of the eddy covariance flux system. All grass leaves were separated from the stem and litter components, and the green areas were measured by running them through a leaf area meter (Li-Cor 3100, Lincoln, NE, USA) (Ma et al., 2017).

3. Methods

Fig. 4 shows schematically how various RT models were coupled in order to retrieve and map vegetation properties during this drought episode from Landsat TOA radiance data. There were four main steps in our study. First, we made use of MODTRAN to obtain the best fitting atmospheric properties over a lake nearby the study site. This resulted in finding the best visibility and aerosol model for various conditions by means of a look-up table (LUT). Second, we ran MODTRAN with the retrieved best fitting properties of the atmosphere beside the other needed inputs one more time to generate the time series of atmospheric transfer functions (for all overpass days). In parallel and as the third step, we used the BSM model to scale the dry soil spectrum using measured soil moisture contents during the episode in order to obtain realistic soil reflectance spectra over time. This scaled soil reflectance value was used as an input besides the other needed parameters for RTMo to generate surface reflectance factors. Fourth, we coupled the MODTRAN generated atmospheric transfer functions with the RTMo generated surface reflectance factors to simulate Landsat TOA radiances during the episode. This assisted in mapping vegetation properties through model inversion by means of numerical optimization. We explain all the applied models with sufficient details including the needed input values in the following sections.

3.1. Radiative transfer models

3.1.1. MODTRAN model

MODTRAN is an atmospheric RT model and the successor of the LOWTRAN model (Kneizys et al., 1988). MODTRAN includes a detailed spectral database of solar irradiance plus the absorption of all important

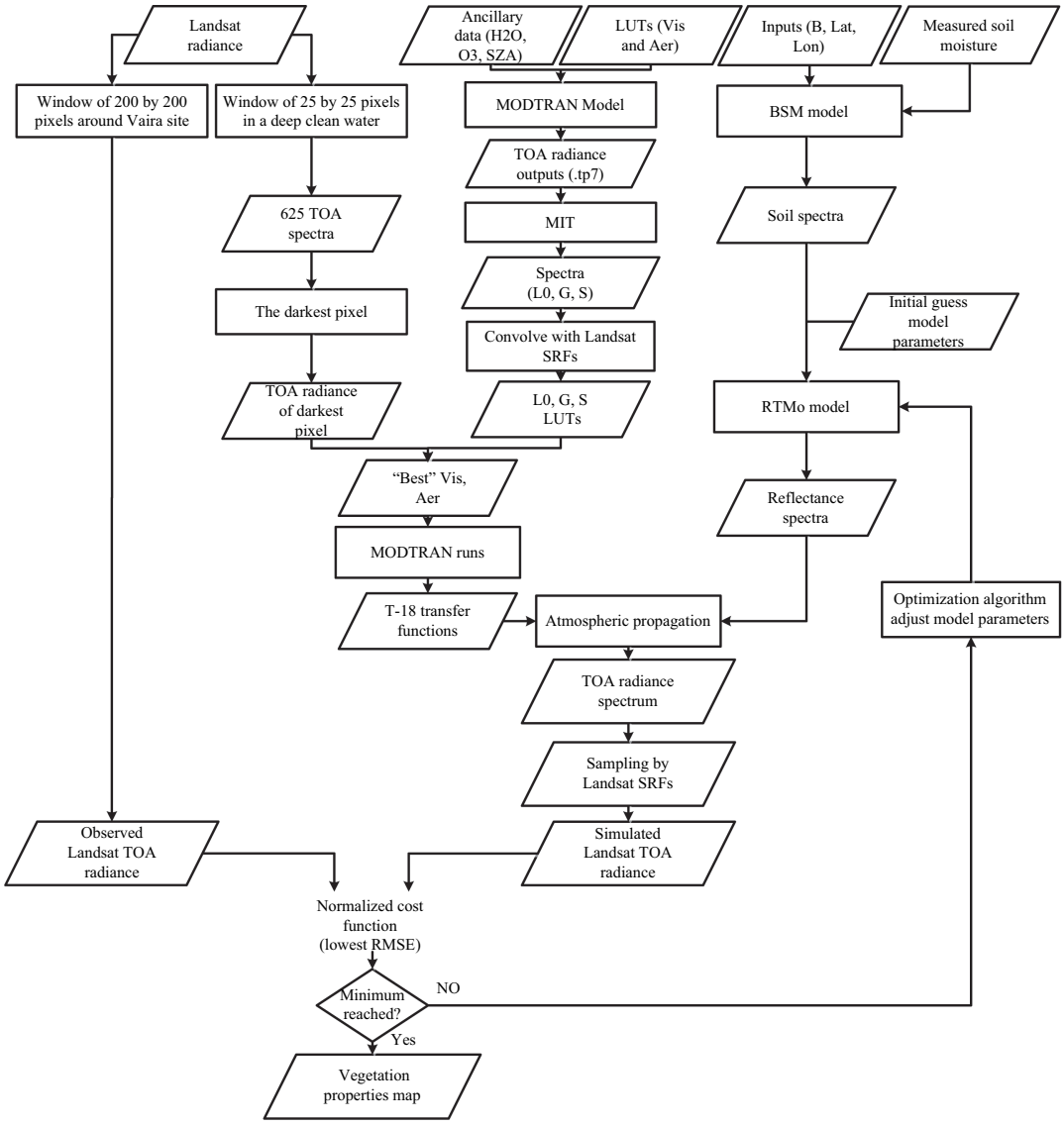


Fig. 4. Flowchart of the TOA approach used in this study to show how various RT models were coupled to retrieve and map vegetation properties during the drought episode from Landsat TOA radiance data.

Table 1
Input parameters used to construct separate LUTs in this study for MODTRAN 5 simulations.

Parameter	Unit	Value/Range	Parameter Status
Aerosol Model	–	Rural, Maritime, Urban	Varied (per image)
Visibility	km	5–100 (5 km increment)	Varied (per image)
Atmospheric profile	–	Mid Latitude Summer	Fixed
DISORT number of streams	–	8	Fixed
Concentration of CO_2	ppm	390	Fixed
Water vapor column	g cm^{-2}	Appendix A.	Fixed
Ozone	DU	Appendix A.	Fixed
Sun zenith angle	degree	Appendix A.	Fixed
Sun azimuth angle	degree	Appendix A.	Fixed
Surface height	m	129	Fixed
Sensor Height	km	705	Fixed
Molecular band model resolution	cm^{-1}	1	Fixed
Start, ending wavelength	nm	400–2400 (optical)	Fixed

atmospheric gases at a fine spectral resolution (Verhoef and Bach, 2003b). In our study, we used MODTRAN version 5.1.2 (Berk et al., 2008). To run MODTRAN, several input parameters describing the real atmospheric and sensor geometric conditions should be prepared (Table 1). In this study we used MODTRAN mainly for two purposes: first to construct a Look up Table (LUT) by varying the aerosol types (3 cases as rural, maritime and urban) and visibility (5 to 100 km with a step of 5 km) for all days (Table 1) to compute surface reflectance over a water body. For every Landsat image a separate input file was prepared by adjusting the local atmospheric conditions, geometric variables and a LUT containing 60 scenarios for various visibility and aerosol models for that specific run of MODTRAN. This resulted in obtaining the best combinations of visibility and aerosol model for all days during this drought episode. Second, we executed MODTRAN for each day separately by preparing an input file including the best visibility and aerosol model (obtained in the previous step) with two runs at the bottom of atmosphere for albedo values of 0.5 and 1, and two runs at TOA for albedo values of 0 and 1. This resulted in obtaining the so-called T-18 atmospheric transfer functions (Verhoef et al., 2018), using the latest version of the MODTRAN Interrogation Technique (MIT) (Verhoef and Bach, 2012) to convert reflectance factors simulated by the RTMo

canopy model to TOA radiances. We obtained one set of T-18 transfer functions for each day and 24T-18 files in total during the episode. Table 1 presents the most important inputs used for such MODTRAN runs.

The atmospheric water vapor column (H_2O) data were obtained from the NASA Giovanni platform (http://giovanni.sci.gsfc.nasa.gov/), the ozone (O_3) data obtained from NASA ozone products (https://ozoneaq.gsfc.nasa.gov/data/ozone/). Geometric variables (i.e., sun zenith angle [SZA] and sun azimuth angle [SAA]) were extracted from satellite image metadata. It is shown that the difference between calculated geometric variables at the Vaira site (representing a pixel) and those extracted from the image metadata (representing the whole image) was quite small for a specific day at the time of satellite overpass (see Appendix A).

In the first phase of the MODTRAN simulations, to obtain the best visibility and aerosol model above a lake, the MIT was applied for each scenario by using surface albedo values of 0.0, 0.5 and 1.0 (Verhoef and Bach, 2003b). The MODTRAN .tp7 output file can be used in the MIT to derive three relevant MODTRAN parameters [i.e., path radiance (L_0), gain factor (G) and spherical albedo (S)]. These three parameters are spectrally variable and changing with atmospheric conditions (Verhoef and Bach, 2003b). The spectral response functions (SRFs) of the Landsat bands were convolved with spectra of the MODTRAN parameters to compute the mean path radiance, gain factor and spherical albedo for every Landsat band in the optical domain.

To collect path radiance, gain factor, and spherical albedo, we need mainly two outputs of the MODTRAN .tp7 file. These are called "SOL_SCAT" (the solar multiple-scattered radiance term of the path radiance) and "GRND_ RFLT" (the total ground reflected radiance).

After obtaining G, L_0, and S for all cases in the LUT, we employed deep water bodies present in the study area and computed water reflectances (at bottom of atmosphere) for all cases in the LUT as follows (Berk et al., 2008; Verhoef and Bach, 2012):

$$R = \frac{L^{\mathrm{TOAm}} - L_0}{G + S\,(L^{\mathrm{TOAm}} - L_0)} \tag{1}$$

where R is the water reflectance, L^{TOAm} is the TOA radiance ($\mathrm{Wm^{-2}\,sr^{-1}\,\mu m^{-1}}$) observed by the sensor for the darkest pixel in a window of 25 by 25 pixels extracted from the deepest part of the clear water of the biggest nearby lake, located in the south part of the image and with a distance of about 25 km from the Vaira site.

We retrieved the best atmospheric properties (L_0, G and S) and, therefore, obtained the corresponding visibility and aerosol model, by selecting the path radiance which gave a reflectance closest to zero from the LUT, since we assumed that the reflectance of clear water is negligible in all visible bands.

To collect the T-18 atmospheric functions, we need seven outputs of the MODTRAN .tp7 file called "TOT_TRANS" (total transmittance), "PTH_THRML" (thermal emission), "SURF_EMIS" (surface emission directly transmitted to the sensor), "SOL_SCAT" (the solar multiple-scattered radiance term of the path radiance), "GRND_RFLT" (the total ground reflected radiance), "DRCT_RFLT" (the direct solar contribution) and "TOASUN" (top-of-the-atmosphere solar irradiance). The white reference solar radiance spectrum at TOA is placed in T_1, while T_2–T_{14} describe the optical transfer functions of the atmosphere and T_{15}–T_{18} describe four thermal transfer functions of the atmosphere. In our study, we used T_1–T_{14} in order to reproduce the TOA radiances from RTMo simulated TOC reflectance factors considering the anisotropic surface reflection. More details about using these transfer functions are provided in the next section.

3.1.2. RTMo model

The optical radiative transfer routine (RTMo), taken from the SCOPE model (Van der Tol et al., 2009), is basically a combination of the PROSPECT5 leaf model (Jacquemoud and Baret, 1990) and the 4SAIL canopy reflectance model (Verhoef et al., 2007; Verhoef, 1984, 1985). Table 2 shows the input parameters of the RTMo model.

Table 2
Input parameters needed for the RTMo model.

Parameters	Abbr. in model	Unit
Leaf chlorophyll content	C_{ab}	$\mu g\,cm^{-2}$
Leaf water content	C_w	$g\,cm^{-2}$
Leaf dry matter content	C_{dm}	$g\,cm^{-2}$
Senescent material	C_s	–
Leaf area index	LAI	$m^2\,m^{-2}$
Leaf inclination distribution function[a]	$LIDF_a$	–
Bimodality of the leaf inclination[b]	$LIDF_b$	–
Carotenoids	C_{ca}	$\mu g\,cm^{-2}$
Leaf structural parameter	N	–
Hot-spot size parameter[c]	hot	$m\,m^{-1}$
Solar zenith angle	θ_s	deg
Observation zenith angle	θ_o	deg
Relative Azimuth Angle	ψ	deg
Spectra of solar irradiance	E_{sun}	$Wm^{-2}\,\mu m^{-1}$
Spectra of sky irradiance	E_{sky}	$Wm^{-2}\,\mu m^{-1}$

a, b These two parameters of the LIDF (i.e., LIDFa and LIDFb) are utilized to describe the (cumulative) probability distribution function of leaf inclination angles mathematically (Verhoef, 1998). LIDFa describes the average leaf inclination, where a value of −1 indicates the erectophile (mean leaf angle MLA of 81.48°) and a value of +1 indicates the planophile (MLA = 8.52°) distribution, while LIDFb describes the bimodality of the distribution, and is proportional to the standard deviation of the distribution. c this parameter is approximated as the ratio between the average width of the leaves and the canopy height (Verhoef and Bach, 2007).

The anisotropic reflectance factors generated by the RTMo model are coupled with the T-18 atmospheric functions generated by MODTRAN to obtain the TOA radiance as follows:

$$L^{\mathrm{TOAs}} = T_1\left[T_2 + T_8 r_{so} + \frac{(T_9 + T_{14}\overline{r_{sd}})r_{do} + T_{10}\overline{r_{sd}} + T_{11}\overline{r_{dd}}}{1 - \overline{r_{dd}}T_3}\right] \tag{2}$$

where r_{so} is the directional reflectance factor for direct solar incidence (or the bi-directional reflectance factor, BRF), r_{do} is the directional reflectance factor for diffuse incidence, r_{sd} is the hemispherical reflectance factor for direct solar incidence and r_{dd} is hemispherical reflectance factor for diffuse incidence, all simulated by the RTMo model.

The T_i are components of the T-18 atmospheric transfer functions derived from the MODTRAN5 .tp7 output file using the MIT technique: one of the most important ones, T_1 describes the extraterrestrial solar irradiance ($\mathrm{Wm^{-2}\,sr^{-1}\,\mu m^{-1}}$), T_2 the bidirectional atmospheric reflectance at the top of the atmosphere (for a black Earth surface) and T_3 the bi-hemispherical reflectance at the bottom of the atmosphere (spherical albedo). For more details on these atmospheric transfer functions, the reader is referred to Table 3 in (Verhoef et al., 2018). The over bars in Eq. (2) indicate a spatial filtering over the surrounding areas of the target pixel (about 1 km radius, with a kernel that is strongly peaked in the center) to describe the adjacency effect. In this

Table 3
Tuned vegetation properties, their lower boundaries (LB), upper boundaries (UB), initial guess (IG), a priori values (μ) and assumed standard deviation (σ_{p0}) used in this study for the retrieval.

Parameter	Unit	LB	UB	IG	μ	σ_{p0}
C_{ab}	$\mu g\,cm^{-2}$	0	100	40	50	28.8
C_{dm}	$g\,cm^{-2}$	0	0.02	0.005	0.01	0.01
C_w	$g\,cm^{-2}$	0	0.05	0.02	0.025	0.01
C_{ca}	$\mu g\,cm^{-2}$	0	25	10	12.5	7.22
C_s	–	0	1.5	0.1	0.75	0.43
Transformed LAI[a]	$m^2\,m^{-2}$	0	0.7	0.4	Measured	Measured
$LIDF_a + LIDF_b$[a]	–	−1	+1	− 0.5	0	0.58
$LIDF_a - LIDF_b$[a]	–	−1	+1	− 0.2	0	0.58

[a] These parameters are transformed to new ones (see Section 3.2.2).

study, we ignored the over bars and used the output of RTMo (r_{sd} and r_{dd}) directly since we assumed that the Landsat pixel was taken from a relatively homogeneous area.

To quantify the surface anisotropy effect and demonstrate its importance, we proposed and computed a novel simple index for the first time, called Anisotropy Index (AI). This index, which is based on the RTMo simulated reflectance factors, is defined as:

$$\mathrm{AI} = \frac{r_{so} - \frac{(r_{do} + r_{dd} + r_{sd})}{3}}{r_{so} + \frac{(r_{do} + r_{dd} + r_{sd})}{3}} \quad (3)$$

The index works similar to a normalized vegetation index (difference over sum), and it can be computed for all wavelengths. It should be noted that the second terms in Eq. (3) compute the average of three reflectance factors. In this study, we computed the AI from 400 to 2400 nm. In the case of a Lambertian surface, the anisotropy index would be zero at all wavelengths, since all four reflectance factors are equal for a Lambertian surface. However, if the surface is non-Lambertian, which is usually the case for vegetation canopies, then the reflectance factors are not equal anymore and, therefore, the anisotropy index would not be zero. Due to the fact that most canopies are non-Lambertian, it is crucial to differentiate among these four reflectance factors and consider the anisotropy effect in the RT modeling.

3.1.3. BSM model

To consider the effects of the soil background on the simulated spectra, we incorporated the BSM soil reflectance model. The BSM model is based on the library of global soil vectors (GSV) proposed by (Chongya and Hongliang, 2012), extended with a brightness-shape transformation and a soil moisture effect to describe soil spectra. The model contains the following parameters: (1) the soil brightness parameter, *B* (dimensionless); (2) the spectral shape "latitude", *Lat* (deg); (3) the spectral shape "longitude", *Lon* (deg); and (4) the soil's moisture volume percentage, *SM*p (%). This model can be used as an alternative for providing a measured soil's spectrum as input. However, if the dry soil reflectance is available, which was the case in our study, one can invert the BSM model and derive the BSM parameters (*B*, *Lat* and *Lon*) from the given soil reflectance spectrum. In the BSM model, one can also scale the dry soil spectrum by the measured soil moisture value and produce realistic soil reflectance spectra for wet soils up to 55% soil moisture content in the top soil layer. For more details on the BSM model, the reader is referred to (Verhoef et al., 2018).

3.2. Parameter retrieval from TOA radiance spectra

3.2.1. An overview of the algorithm

The observed TOA radiance by Landsat can be compared to the simulated TOA spectra by the coupled RT model. We used Numerical Optimization (NO) to invert the coupled model, similar to (Bayat et al., 2016a; Van der Tol et al., 2016; Verhoef et al., 2018), against the time series of Landsat-observed TOA radiance and, consequently, to retrieve one set of vegetation properties separately for each of the observation days during the episode. We ran the model in MATLAB iteratively, with specific vegetation properties (see Table 3) in order to minimize a normalized cost function *C* as:

$$C = \sum_{i=1}^{Nbands} \left(\frac{L_i^{\mathrm{TOAm}} - L_i^{\mathrm{TOAs}}}{\sigma_i} \right) + \sum_{j=1}^{Npar} \left(\frac{p_j - u_j}{\sigma_{p_0 j}} \right) \quad (4)$$

where L^{TOAm} is the Landsat-observed TOA radiance ($\mathrm{Wm^{-2}\,sr^{-1}\,\mu m^{-1}}$), the L^{TOAs} is the model simulated TOA radiance ($\mathrm{Wm^{-2}\,sr^{-1}\,\mu m^{-1}}$), the σ is uncertainty (i.e., standard deviation) of the observed TOA radiance (see Section 3.2.4), i indicates the Landsat optical band number (ranges from 1 to 6), the p are the current and μ are the priori values of the vegetation properties, the σ_{p0} is the assumed uncertainty of the a priori values of each property (see Section 3.2.3) and the j indicates the 7 retrieved parameters (Table 3).

We made use of the MATLAB built-in function "lsqnonlin" from the optimization toolbox, selecting a Trust Region algorithm in order to update the values of properties after each iteration step. The stopping criterion for the iteration was fulfilled when the improvement of the cost function was less than 10^{-3}, similar to (Van der Tol et al., 2016; Vilfan et al., 2016). The starting values, also called the initial guess (IG), the a priori values (μ) and their uncertainties (σ_p) of the input parameters for the retrieval algorithm are presented in Table 3.

The choice of values for LB, UB and IG were adapted from the previous studies (Bayat et al., 2016a; Van der Tol et al., 2016; Verhoef et al., 2018). Although the IG is necessary for NO as the starting point, the choice of their values is subjective but does not affect the results (Bayat et al., 2016a; Vilfan et al., 2016). The use of uncertainties [i.e., standard deviations (σ_{p0} and σ)] to normalize the cost function and the a priori values is an effective regularization approach applied in NO (Dorigo et al., 2007; Verhoef et al., 2018) to reduce numeric ill-posedness [i.e., multiple solutions yielding the same spectrum (Combal et al., 2003)], since in the case of ill-posedness the solution is automatically driven in the direction of the a priori values. In addition, for the *N* and *hot* parameters we considered fixed values of 1.5 and 0.05 based on previous studies (Bayat et al., 2016a; Haboudane et al., 2004; Houborg et al., 2007). The dry soil parameters for the BSM soil reflectance model were obtained by their retrieval from a nearby bare soil pixel (as $B = 0.55$, $Lat = 19.50$ and $Lon = 49.46$) at the end of the episode when the soil was completely dry. For the rest of the time series, we used the measured surface soil moisture values collected on various days as a priori information to scale the dry soil spectra for moist conditions during the episode and, therefore, to take into account the soil moisture effects on dry soil spectra, keeping *B*, *Lat* and *Lon* at the above mentioned values.

3.2.2. Transformed parameters

Some input parameters in RTMo were transformed in the inversion process, notably *LAI*, $LIDF_a$ and $LIDF_b$. *LAI* was transformed to a new parameter $1 - \exp(-0.2 \times LAI)$ to improve the linearity of the model. The value of 0.2 ensures a more linear response of the reflectance in the near-infrared part, which is the most sensitive part of the spectrum to high values of *LAI* (Verhoef et al., 2018). Moreover, *LIDF* parameters were transformed because $LIDF_a$ and $LIDF_b$ are restricted by the condition $|LIDF_a| + |LIDF_b| \leq 1$. By constructing two transformed variables, via summation ($LIDF_a + LIDF_b$) and subtraction ($LIDF_a - LIDF_b$) and attaching -1 and $+1$ as their LB and UB, one can ensure that the constraint of $|LIDF_a| + |LIDF_b| \leq 1$ is effectively fulfilled in an automatic way (Verhoef, 1998).

3.2.3. Including the a priori information and uncertainties

For the a priori values, we used measured *LAI* information during this drought episode for the case of *LAI* in the retrieval. This is useful to constrain the inversion and to reduce the volume of parameter space. However, we did not know any a priori values for the other properties (C_{ab}, C_w, C_{dm}, C_s and *LIDF*) on the ground during the episode. Thus, for these cases, we used the mid of the total range [i.e., (LB + UB) / 2] as the a priori values, similar to (Verhoef et al., 2018). Furthermore, for the a priori uncertainties, we assumed uniform priori distributions over this range between LB and UB of the vegetation properties. Assuming uniform prior distributions over the range between LB and UB produced values of uncertainty equal to $1/\sqrt{12} \approx 0.3$ times the range of each property. Moreover, we used measured SM values as a priori information to scale the dry soil spectra for moist conditions during this drought episode.

3.2.4. Sensor noise

We considered the digitization noise as a rough estimate of sensor noise and, therefore, the source of uncertainty for Landsat-observed TOA radiances. It is necessary to obtain reasonable estimates for sensor

Table 4
Digitization noise in Landsat TM5 and ETM7 and assumed standard deviation (σ) used in this study for different bands.

Landsat bands	Digitization noise [$Wm^{-2}sr^{-1}\mu m^{-1}$]		σ [$Wm^{-2}sr^{-1}\mu m^{-1}$]	
	TM5	ETM7	TM5	ETM7
Band 1	0.766	0.779	0.230	0.234
Band 2	1.448	0.799	0.434	0.240
Band 3	1.044	0.622	0.313	0.186
Band 4	0.876	0.640	0.263	0.192
Band 5	0.120	0.126	0.036	0.038
Band 7	0.066	0.044	0.020	0.013

noise in the different Landsat bands, and one may assume that digitization noise in each band has been designed to be in proportion to actual sensor noise. Maximum digitization noise may be considered equal to 1 DN count in each band of Landsat, because the maximum amplitude of the coherent noise in Landsat sensor is equal to 1 DN (USGS, 1998). Consequently, assuming uniform (a priori) distributions over a range of 0–1 DN leads to the uncertainty of 0.3 DN for each band of Landsat. Thus, we converted 1 DN in each band of Landsat into TOA radiance values and computed their uncertainties as shown in Table 4.

3.2.5. Error propagation analysis

The propagation of sensor noise in the Landsat-observed TOA radiances into errors in the retrieval of vegetation properties is estimated by:

$$\sigma_P = (J_L{}^T J_L)^{-1} J_L{}^T \sigma_i \quad (5)$$

The σ_p is the uncertainty in the vegetation properties propagated from the sensor noise (σ_i) and J_L is the Jacobian of the model computed for each set of optimized vegetation properties expressed by:

$$J_L = \begin{pmatrix} \partial L_{\lambda 1}/\partial_{P1} \cdots \partial L_{\lambda 1}/\partial_{P7} \\ \cdots \quad \cdots \quad \cdots \\ \partial L_{\lambda 6}/\partial_{P1} \ldots \partial L_{\lambda 6}/\partial_{P7} \end{pmatrix} \quad (6)$$

The Jacobian J_L is a matrix containing the first partial derivatives of the relative model output (Bowyer and Danson, 2004; Saltelli et al., 2000). We changed vegetation properties by a small step (10^{-6}) and recorded the response in the simulated TOA radiances to such a small change. This is basically the local sensitivity analysis of the radiance L at each wavelength λ_i of the 6 Landsat optical spectral bands (rows, Table 4) to each of the 7 input parameters (columns, Table 3). The computed Jacobian was then utilized to relate the observed spectra (Table 4) to uncertainty in the retrieved properties (Fig. 17).

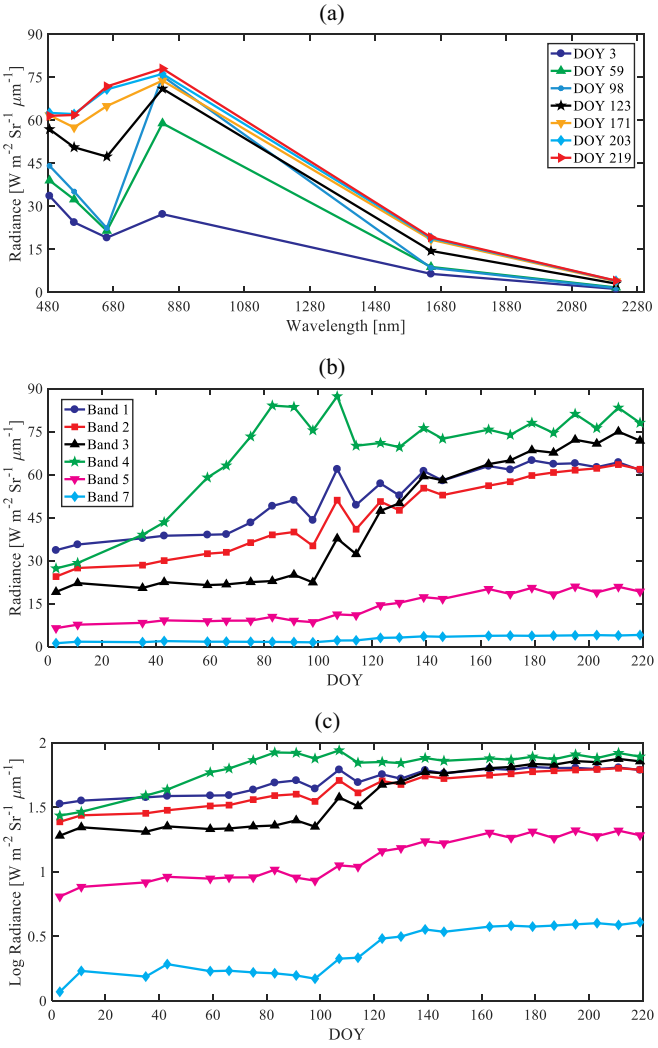


Fig. 6. Time series of Landsat TOA radiance spectra observed at the Vaira site during the 2004 drought episode. Radiance variations observed on (a) different days as a function of wavelength and (b) different bands as a function of time (DOYs). The logarithms of Landsat TOA radiance variations are shown (c) as a function of time (DOYs) to better detect the relative changes.

4. Results

4.1. Landsat observed TOA radiance variations over time

Three representative examples (on DOYs = 59, 75 and 203) of the

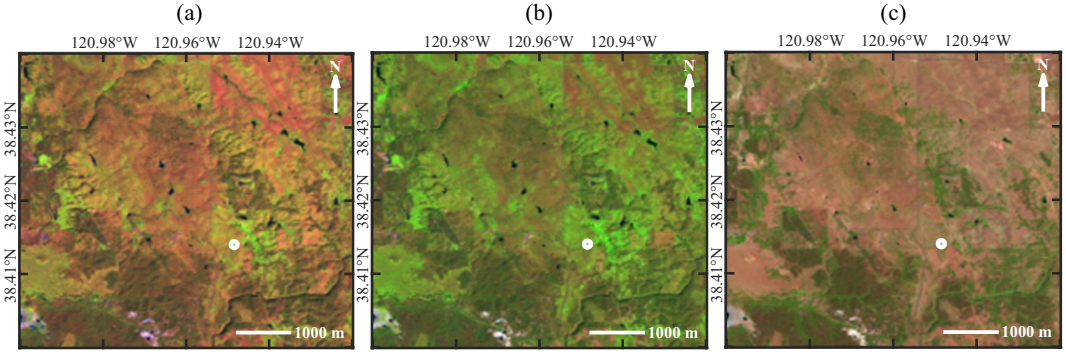


Fig. 5. TOA radiance images observed by Landsat TM5 (red = band 5, green = band 4, blue = band 3) for (a) DOY 59 (SM = 26%) (b) DOY 75 (SM = 19%) and (c) DOY 203 (SM = 2.2%) in 2004. The white circle shows the location of the Vaira site. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

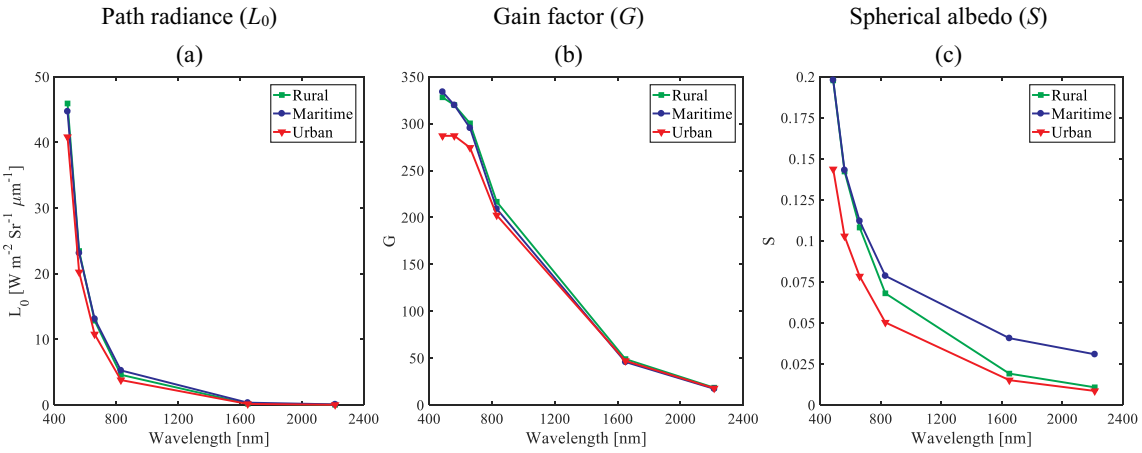


Fig. 7. MODTRAN parameters spectral variations at a fixed visibility of 25 km over various aerosol types over the Vaira site for DOY 107 in 2004. (a) Path radiance (L_0), (b) Gain factor (G) and (c) Spherical albedo (S) for DOY 107 in 2004.

TOA radiance observed by Landsat (in various soil moisture conditions) during this drought episode are presented in Fig. 5. These images clearly highlight that not only the vegetation properties are changing significantly during the developing drought episode, but that the background soil contribution varies over time as well. For instance, there is more contribution from the vegetation at DOY 75 (the middle of the episode (Fig. 5b)), whereas there is more contribution from the soil at DOYs 59 (early in the episode (Fig. 5a)) and 203 (the end of the episode (Fig. 5c)).

To better understand the trends in the observed radiances during this drought episode, we analyzed the radiance changes at the Vaira site over time (Fig. 6). The radiance in bands 1, 2, 3, 5 and 7 increased over time (Fig. 6b, c). However, the radiance in band 4 initially increased considerably (until DOY 80), decreased in the middle period (DOY 80–112) and was almost stable afterwards (≥DOY 112) (Fig. 6b, c). Radiance increase of the red band (band 4) was the strongest, the difference between the radiance spectra at the highest (DOY 3) and the lowest soil moisture (219) was significant for almost all bands (Fig. 6a). All bands showed mid-event fluctuations (between DOYs 90–120).

4.2. MODTRAN model

4.2.1. MODTRAN parameter variations

MODTRAN parameters (i.e., L_0, G and S) resampled to the Landsat bands shown for various atmospheric aerosol types (rural, maritime and urban) for a fixed visibility of 25 km on 16 April 2004 (DOY 107) are presented in Fig. 7. The atmospheric L_0 shows the case when there is no reflectance from the surface and, therefore, the total radiance comes from atmospheric scattering alone. As Fig. 7 shows, L_0 follows a downward trend with wavelength in the solar reflective domain. This is due to the fact that the atmosphere scatters much more at shorter wavelengths (band 1) than at longer wavelengths (band 7). The G, which includes the extraterrestrial solar irradiance and the total two-way transmittance in the atmosphere, roughly follows the shape of the solar irradiance spectrum. However, similar to the L_0, the S shows a smooth downward trend over wavelength. The L_0, G and S variations are not only a function of wavelength and visibility, but also of the aerosol type. In general, the urban aerosol model had a stronger absorption and, therefore, in all cases showed lower values compared to the other two aerosol models. Moreover, the L_0, G and S variations show a stronger spectral slope for the rural case compared to the maritime.

The variations of the L_0, G and S on 16 April 2004 (DOY 107), for a range of haze conditions (i.e., visibilities of 5, 15, 30 and 100 km) and three aerosol models (rural, maritime and urban) are presented in Fig. 8.

As Fig. 8a–c shows, the atmospheric L_0 for the rural (Fig. 8a) and maritime (Fig. 8b) aerosol cases had higher ranges of variation than that of the urban aerosol case (Fig. 8c). In addition to the dependency of the L_0 on aerosol types, the results show the effects of visibility: the lower visibility, the higher L_0, and vice versa. Fig. 8d–f show that the G in rural (Fig. 8d) and maritime (Fig. 8e) aerosol had a higher range of variations compared to that of urban aerosol case (Fig. 8f), similar to the atmospheric L_0. In addition, the G spectra in the urban aerosol case respond more strongly to increasing visibilities compared to rural and maritime aerosol types. The G changes directly with visibility conditions: the lower visibility, the lower G, and vice versa. Fig. 8g–i shows that the S in rural (Fig. 8g) and maritime (Fig. 8h) aerosol had a higher range of variations compared to that of urban aerosol case (Fig. 8i), similar to the atmospheric L_0. Moreover, the S spectra in the rural and maritime aerosol case, corresponding to various visibilities showed larger deviations compared to the urban aerosol type. In addition to the dependency of the S to aerosol types, the results show again the effect of visibility: the lower visibility, the higher S and vice versa.

A wrong assumption on visibility or aerosol type will produce a wrong computation of the TOA radiances and consequently, vegetation properties retrieval will be biased, especially for urban (G, S), rural (S) and maritime (S) cases.

4.2.2. The best fitting atmospheric properties

We compared the surface reflectance generated by the retrieved best atmospheric properties (MODTRAN generated reflectance) over a lake with the Landsat surface reflectance products obtained from USGS ESPA on-demand system (http://espa.cr.usgs.gov/) at the same dates (as for Landsat radiance data in Fig. 3) in our study site.

The results demonstrated that the extracted values from the USGS Landsat surface reflectance products were relatively far from zero over a clear water. Although slightly positive values are acceptable, negative reflectance values are definitely due to atmospheric correction errors in the USGS products and will cause problems in quantitative analysis. Three representative examples of the reflectance values extracted by a window of 5 by 5 pixels from USGS Landsat products over a deep water body on various days are presented in Table 5. Our MODTRAN best fitting parameters were forced to generate reflectances of zero over the same lake on the same dates.

4.2.3. T-18 atmospheric transfer functions

Fig. 9 presents two representative examples of the spectra of the generated functions T_2–T_{14} over the wavelength range of 400–2400 nm for DOY 66 (Fig. 9a, Vis = 65 km and aerosol type = urban) and DOY 107 (Fig. 9b, Vis = 25 km and aerosol type = maritime).

Although almost similar spectral shapes were obtained for the

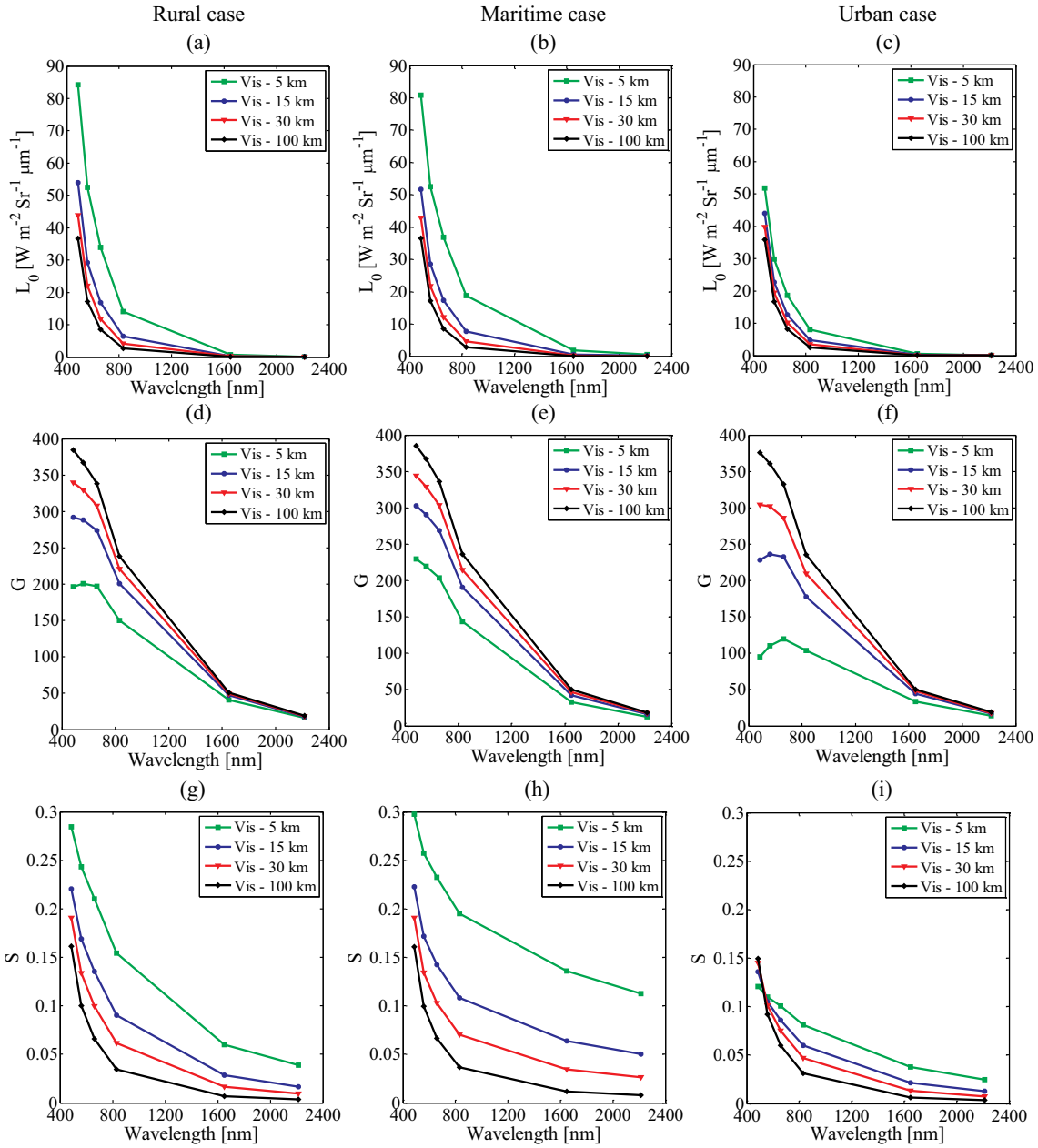


Fig. 8. MODTRAN parameters (i.e., path radiance (L0), gain factor (G) and spherical albedo (S)) spectra variations at various visibilities (5, 15, 30 and 100 km) and aerosol types (rural, maritime and urban cases) for DOY 107 at Vaira site. Left panels (a, d, g) show the rural case. Middle panels (b, e, h) show the maritime case and right panels (c, f, i) show the urban case.

Table 5
The reflectance in various bands extracted by a window of 5 by 5 pixels from USGS Landsat surface reflectance products over a water body during this drought episode for (a) DOY 59, (b) DOY 75 and (c) DOY 203 in 2004.

Landsat bands	DOY 59			DOY 75			DOY 203		
	Darkest pixel	Window mean	Window std	Darkest pixel	Window mean	Window std	Darkest pixel	Window mean	Window std
Band 1	0.028	0.034	0.003	0.028	0.031	0.003	0.023	0.027	0.002
Band 2	0.009	0.019	0.004	0.014	0.020	0.003	0.014	0.015	0.002
Band 3	0.012	0.014	0.004	0.008	0.013	0.002	0.007	0.009	0.001
Band 4	0.011	0.012	0.002	0.011	0.013	0.002	0.008	0.011	0.001
Band 5	−0.005	0.001	0.003	−0.004	0.003	0.002	−0.001	0.003	0.002
Band 7	−0.007	0.001	0.004	−0.001	0.002	0.003	−0.001	0.003	0.002

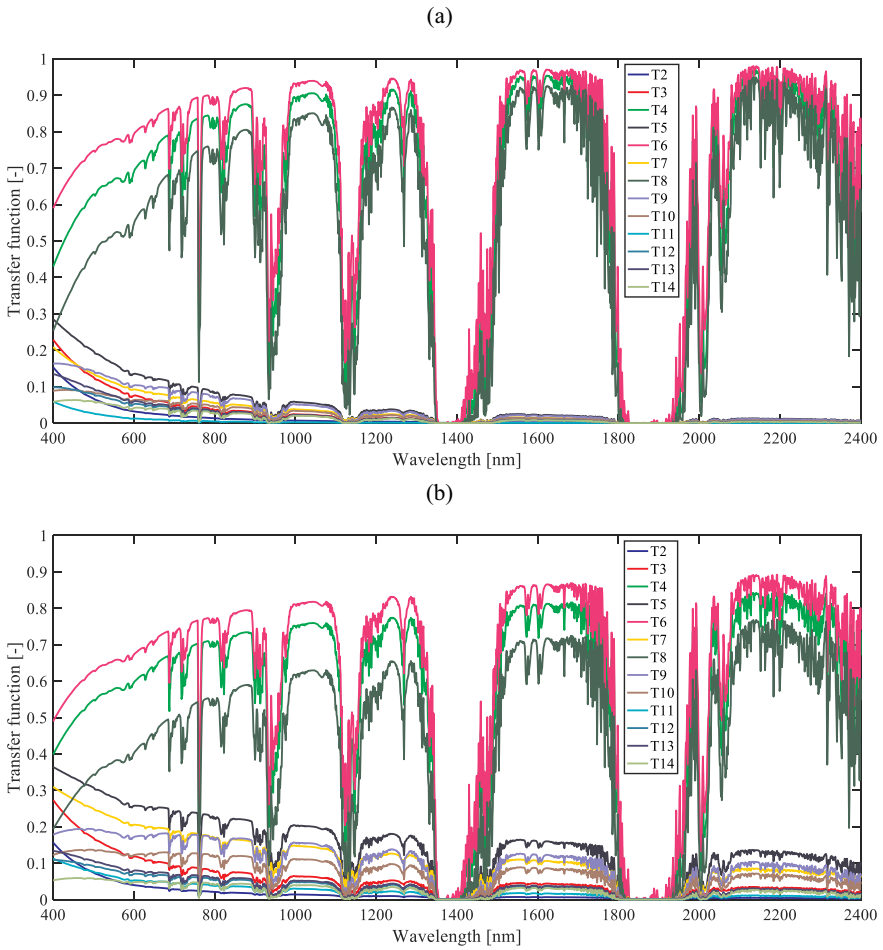


Fig. 9. Atmospheric spectral transfer functions T_2–T_{14} generated for (a) DOY 66 (Vis = 65 km and aerosol type = urban) and (b) DOY 107 (Vis = 25 km and aerosol type = maritime) in 2004 at the Vaira site.

T_2–T_{14} system for DOYs 66 and 107, their magnitudes and range of variations were clearly different. The lower visibility on DOY 107 gives lower transmittances and higher scattering, which is confirmed by the lower levels of functions T_4, T_6 and T_8, and the higher levels of the other T-functions on DOY 107.

4.3. Coupled model inversion against TOA radiance

Three representative examples of simulated TOA radiance images by the coupled model and the residual maps of spectral fitting (i.e., the differences between the Landsat-observed TOA radiance and the simulated TOA radiance) are presented in Fig. 10.

The smallest error (RMSE) between observed and simulated TOA radiance was used as a measure of the model inversion performance. Fig. 11 illustrates the time series coupled model fittings at the Vaira site obtained through the optimization, between observed spectra (by Landsat) and simulated spectra (by model) in a window of three by three pixels around the flux tower at the site on various days. Overall, the small error in the model inversion ($0.006 < \text{RMSE} < 2.2\,\text{Wm}^{-2}\,\text{sr}^{-1}\,\mu\text{m}^{-1}$) shows that the model was able to accurately reproduce the observed optical spectra under different soil moisture conditions during the whole observation period.

As discussed in Section 3.2, we scaled the dry soil spectrum based on the measured surface soil moisture and used them in the RTMo model. Fig. 12 illustrates the fixed dry soil spectrum in red and the scaled soil reflectance for various days at the Vaira site in blue dashed lines. These results showed that the dry soil spectrum provides a realistic case for DOYs ≥ 98.

Fig. 13 illustrates the time series of computed anisotropy index spectra at the Vaira site on various days during the episode. Considering the variation of the computed index ($-0.1 < \text{AI} < 0.28$), it is clear that the grass canopy is non-Lambertian and, therefore, it is relevant to consider the anisotropy effect for simulating surface reflectance, TOA radiance and consequently for the retrieval of vegetation properties. As the anisotropy index spectra show, canopy anisotropy has to be taken into account in all parts of the spectrum during the whole episode from normal condition to severe stress. However, the largest wavelength-specific effects of canopy anisotropy are observed from DOY 3 to DOY 123 (during the near normal and mild stress period). So, anisotropy appears to reduce under dry conditions at this grassland site. It should be noted that the anisotropy index also reveals two prominent peaks in the short wave infrared region (~1400 nm, ~1900 nm) from DOY 3 to DOY 146.

We present the anisotropy index variations for all DOYs together in Fig. 14. In general, from the middle of the episode, when soil moisture and vegetation coverage decreased, the surface anisotropy effect decreased as well. Considering that the RTMo model includes the assumption of a Lambertian soil background, it is not surprising that the modeled anisotropy decreases when more soil becomes exposed.

We compared the retrieved *LAI* values with those of measured *LAI* collected during this drought episode (Fig. 15). The model inversion gives a prediction of *LAI* within the measured *LAI* ± the uncertainties. We expected the retrieved values to match with measured *LAI* well since the measured *LAI* was used to constrain the model inversion. Except for very low LAI values, the results fulfilled our expectations.

However, overestimation of *LAI* at the end of the episode (for lower

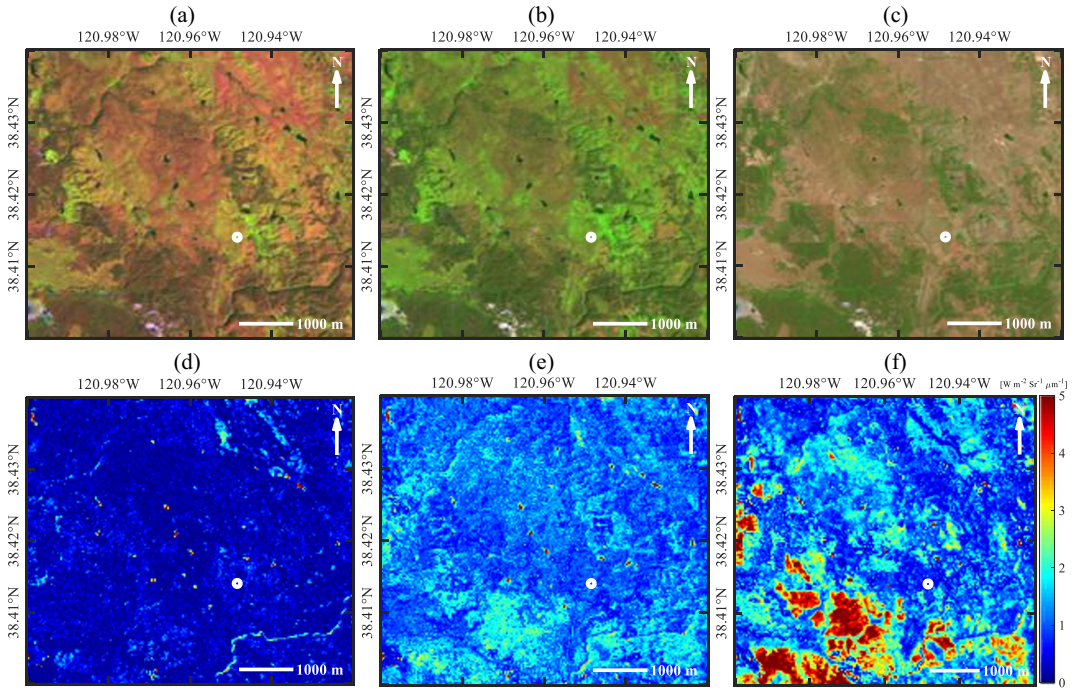


Fig. 10. Simulated TOA radiance images (red = band 5, green = band 4, blue = band 3) by RTMo model for (a) DOY 59, (b) DOY 75, (c) DOY 203 in 2004 and the residual maps of spectral fitting (i.e., the differences between the observed TOA radiance and the simulated TOA radiance over the spectra) for (d) DOY 59, (e) DOY 75, (f) DOY 203. The white circle shows the location of the Vaira site. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

LAI values) can be attributed to the fact that the measured *LAI* only refers to the green *LAI* and, therefore, the brown leaves inside the sample plots are ignored. In the RTMo model, only the effective (total) *LAI* is considered, which results in higher values of *LAI* at the end of episode where fewer green leaves, but still many brown leaves, are present.

4.4. Retrieved properties variations during the episode

4.4.1. Spatio-temporal changes

Maps of the retrieved vegetation properties were generated for all the observations during the developing drought episode. Three representative examples of the retrieved maps for three days (DOY 59, 75 and 203) during the episode are presented in Fig. 16.

To better understand the trend of retrieved vegetation properties during the drought episode, we analyzed their changes over time at the Vaira site (Fig. 17). *LAI*, $C_{ab,}$ and C_w showed an initial increase in normal conditions to the middle of mild stress period (DOY 90) during the episode, but a decrease afterwards (from the middle of mild stress period to severe stress). Maximum *LAI*, C_{ab} and C_w were retrieved at the middle of the mild stress period, as expected for the normal phenological cycle, while the minimum values for these properties were obtained both in the beginning (the normal period) and during the later severe stress period due to drought.

C_{dm} and C_s showed the opposite trend compared to *LAI*, C_{ab} and C_w: they decreased first, reaching minimal values in the middle of mild stress period and increased later, reaching a maximum at the end of the episode when the stress was severe. The opposite trend of properties between the first group (*LAI*, C_{ab} and C_w) and the second group (C_{dm} and C_s) could be expected. When the vegetation is functioning well (i.e., photosynthesis rates are high), it produces more green leaves resulting in higher *LAI* and, in general high C_{ab} and C_w values, while the opposite is true at the end of episode when the brown leaves are dominant and one can observe higher values for C_{dm} and C_s. Despite a few local fluctuations in $LIDF_a$, it showed an overall declining trend during the early normal period of the episode indicating more erectophile leaves. However, an opposite trend (i.e., upward) in the mild and moderate period of the episode is observed which indicates more planophile leaves. The maximum values of $LIDF_a$ at the end of drought episode indicate that horizontally-oriented (planophile) leaves were dominant, while they were more vertically-oriented (erectophile) leaves in the middle of the episode, albeit with some fluctuations.

As presented in Section 3.2.5, we used the Jacobian *J* of the model to investigate the error propagation of TOA radiance observations into the retrieved vegetation properties. The uncertainty in the retrieved parameters (*LAI*, C_{ab}, C_w, C_{dm}, C_s and $LIDF_a$) originating from the uncertainty in the radiance measurements and propagated to the parameters, is small (error regions in Fig. 17), which could be expected since *LAI* was known from measurements.

5. Discussion

5.1. Time series of TOA radiance observations

The most striking result to emerge from the time series of the observed TOA radiance data is that all spectral bands of Landsat are sensitive to a soil moisture deficit (water stress) over time during the episode (Fig. 5), and therefore investigation of the whole spectrum, by means of RT modeling, is required to exploit the full potential of the spectral information (Bayat et al., 2016a; Carter, 1991; Chávez et al., 2013; Zarco-Tejada et al., 2003). This indicates that, although extracting the information content from parts of the spectrum by means of VIs can assist in detecting stress signals to some extent, it is more advantageous to exploit all the information in the spectra by using the RT modeling approach.

Water stress effects on TOA radiance spectra can be explained from two stages, notably direct effects, also termed primary impacts, and indirect effects, also termed secondary impacts. Direct effects are considered as the impacts of water shortage (dehydration) on vegetation properties related to the spectral properties of water (this gives an

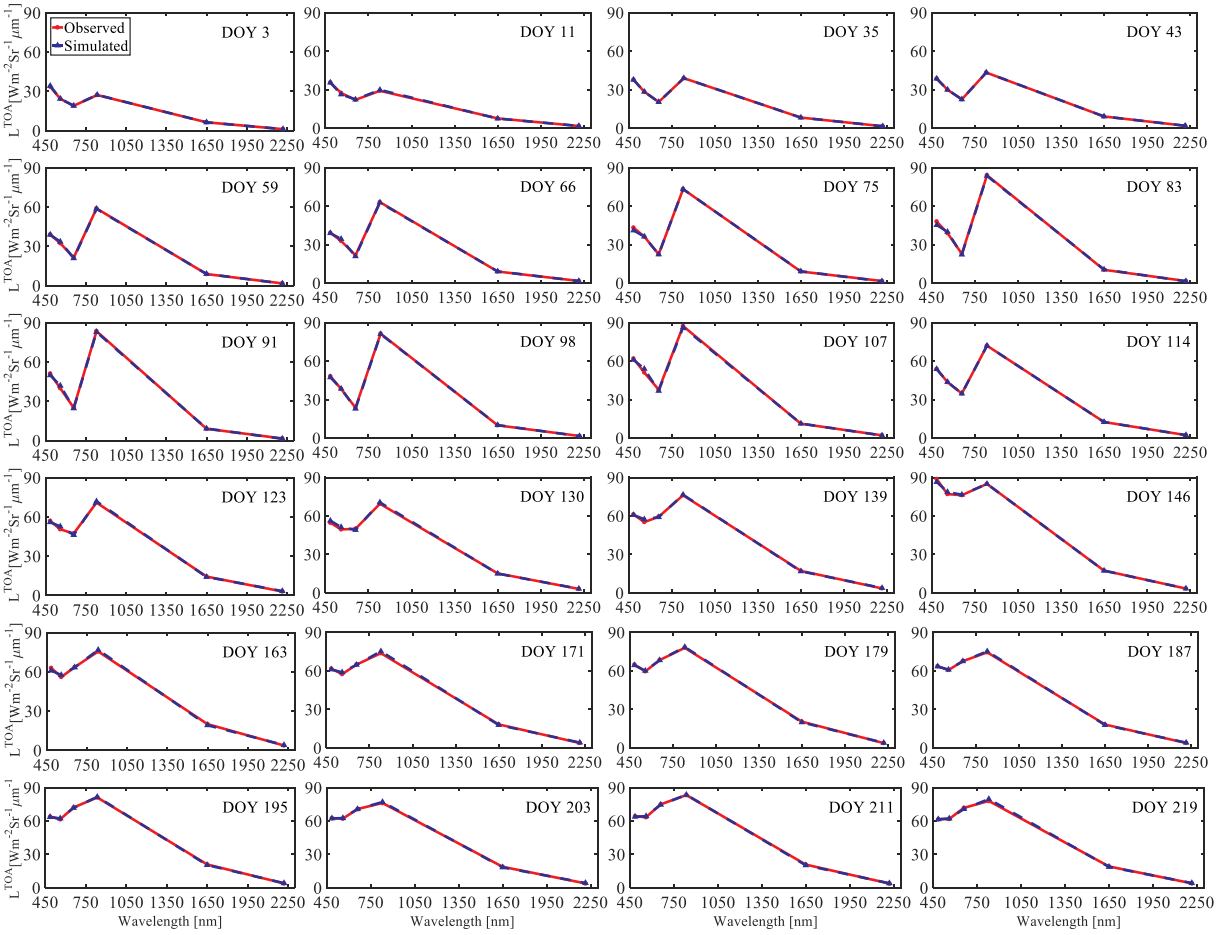


Fig. 11. The goodness of fit for the spectra obtained between observed TOA radiances (at Landsat optical bands; shown as red solid lines) and the simulated TOA radiances (resampled to Landsat optical bands by Landsat SRFs; shown as blue dashed lines) on 24 different days at the Vaira site during the 2004 drought episode. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

increasing reflectance mostly in the short wave infrared (SWIR) part because of less absorption by water). The indirect effects are considered as the impacts that cannot be explained by the spectral properties of water alone, but that are also related to other changes of canopy and leaf structure and the effects of dehydration on pigment concentrations and defoliation. Both types of impacts are observed in our study. Moreover, as Fig. 6 shows, the observations acquired in the first part of the episode show a low radiance in the visible (VIS) and SWIR parts of the spectrum, because of relatively high concentrations of C_{ab} and C_w, respectively. Our findings are in agreement with the findings of previous studies at leaf (Genc et al., 2013) and canopy level (Bayat et al., 2016a; Elmetwalli et al., 2012). Canopy TOA spectral radiance in band 4 of Landsat [i.e., the near infrared (NIR)] initially increased, but in more severe stages of water stress it decreased. Such a pattern could be explained as a result of cell wall deterioration (Knipling, 1970), loss of plant leaves and the changes in the average leaf angle (Asner, 1998). Radiance increases in the VIS parts of the spectrum, i.e., in bands 1–3, could be well explained by C_{ab} reductions due to the water stress. An increase in radiance in the SWIR parts of the spectrum, bands 5 and 7, occurred due to the decline of C_w. Our findings of various spectral changes due to water stress support previous studies (Asner, 1998; Chávez et al., 2013; Clevers et al., 2010; Hunt Jr and Rock, 1989). For detailed discussions about vegetation properties changes, radiance variation, and plant-water relations in various conditions (i.e., stressed and normal), the reader is referred to (Damm et al., 2018).

5.2. Coupling RT models

TOA radiance observations by satellites above a vegetated surface are basically influenced by three main factors: (1) atmospheric conditions, (2) canopy and leaf properties and (3) understory background or soil characteristics (Verhoef and Bach, 2007). Reliable models describing each of these three factors are required to simulate the atmosphere – surface system and to obtain accurate retrievals of vegetation properties. In our study, we considered all of these three contributions in our modeling effort.

We applied the MODTRAN atmospheric, RTMo canopy and BSM soil reflectance models as tools to describe the atmosphere – surface system to simulate TOA radiance observations in a time series and, therefore, to retrieve/map vegetation properties directly. Using this physically-based approach has many advantages, as it is based on physical laws, use of all spectral bands, the least reliance on in situ measurements, and it gives generalizable results (Atzberger et al., 2015; Atzberger et al., 2011). By combining the outputs of MODTRAN to the surface reflectance simulated by the RTMo and BSM models, one can simulate TOA radiance and eliminate the need to perform traditional atmospheric correction. This is significant since even in the best case of atmospheric correction performance, its result is a surface reflectance that is still dependent on atmospheric conditions and the solar zenith angle, unless the surface is both uniform and Lambertian over a large area, but this will never be the case in practice. The retrieval of surface properties by model inversion of TOA radiance data, instead of using surface reflectances obtained after atmospheric correction, is the best

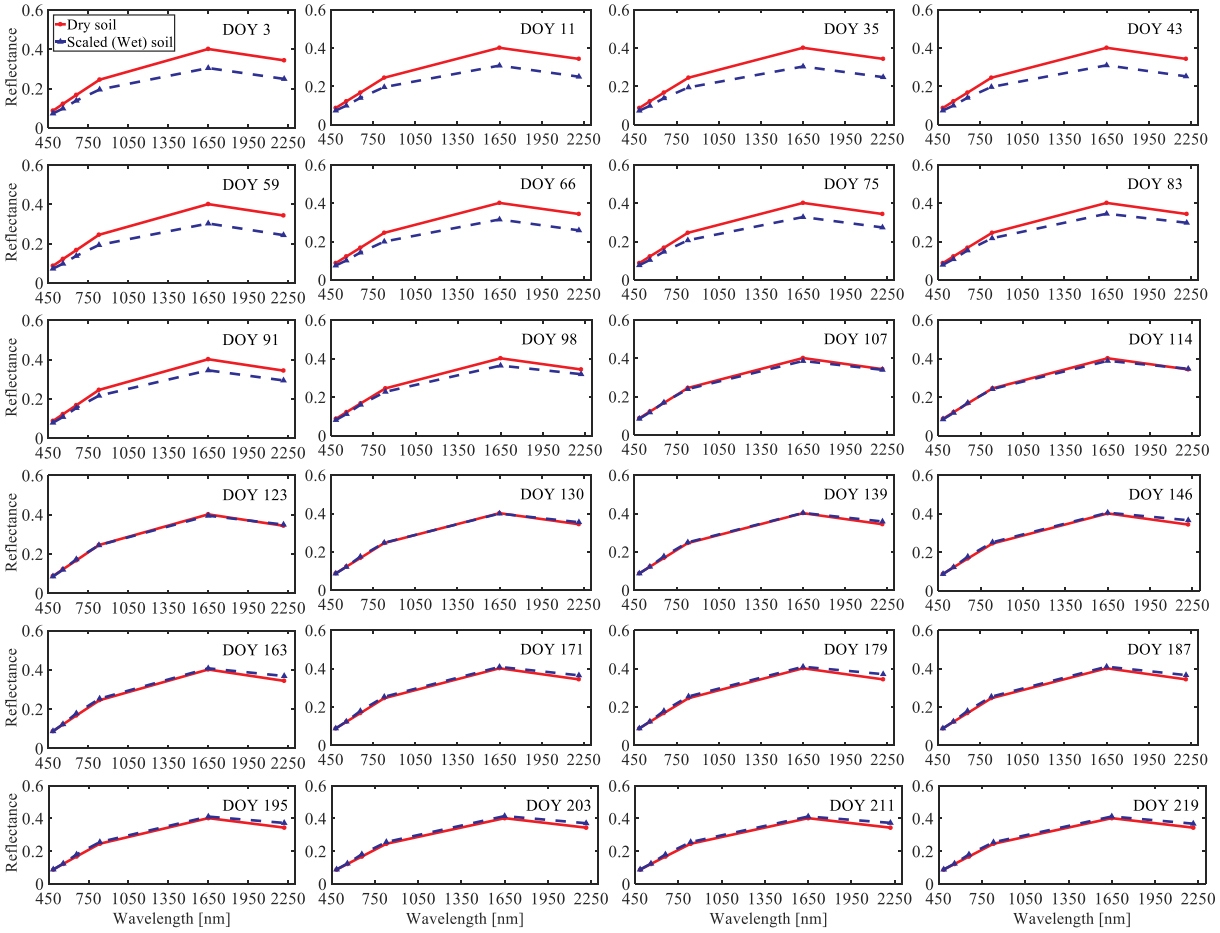


Fig. 12. Soil reflectance simulations with BSM model. The dry soil reflectance (at Landsat optical bands; shown as red solid lines) were scaled to obtain realistic spectra (at Landsat optical bands; shown as blue dashed lines) on different days at the Vaira site during the episode. Dry soil spectrum provides a realistic case for DOYs ≥ 98. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

way to remove systematic effects related to the solar zenith angle and the atmospheric conditions, since RT models can take these effects implicitly into account. Thus, the proposed TOA approach can narrow down the error sources due to the atmospheric correction but also does justice to the actual physics of the satellite observations by simulating the same quantity as observed by the satellite. We used the combination of 3 RT models to retrieve vegetation properties directly from the time series of Landsat images observed during a prolonged drought episode in which soil moisture content declined 10-fold from 0.3 to $0.03\,m^3\,m^{-3}$ during a period of 220 days. This episode provided a suitable case to investigate the performance of MODTRAN – RTMo – BSM, since during this episode both atmospheric (different months) and environmental (drought and soil moisture deficit stress) conditions were affecting the vegetation canopy. We can attribute the success of the combined models to their capability of capturing variations of atmospheric conditions (considering various aerosol types and visibilities) and environmental conditions (considering various soil moisture and vegetation properties) using the Landsat optical bands. Landsat time series have been selected for this research mainly because of their great potential due to the spatial and spectral resolution, the foresighted acquisition and global image archive (Wulder et al., 2012), freely available archived and new data sets (Loveland and Irons, 2016), the continuity of the observations (Loveland and Dwyer, 2012) and the ever increasing range of users and applications (Wulder et al., 2008).

To find out the best fitting atmospheric properties one needs to compare the computed surface reflectance of the LUTs with the ground measured reflectance (as ground truth). Since we did not have such ground truth data for 2004, we employed lakes as dark pixels in the Landsat image near the study site. However, it would be possible to invert the coupled set of MODTRAN – RTMo – BSM models when more spectral data are available (e.g., hyperspectral images) as well. This would eliminate the need for either ground spectral measurements or dark water bodies in the image. The comparison of generated surface reflectance, from the MODTRAN best fitting parameters over a lake, with the USGS surface reflectance products (http://espa.cr.usgs.gov/) over a lake, has further strengthened our confidence in the atmospheric properties retrieval accuracy. We found negative water reflectances in the USGS surface reflectance products over a lake (Table 5), which indeed is due to possible errors in the atmospheric correction and can be attributed to overestimations of the local atmospheric path radiance. Negative reflectance values are physically impossible and can no longer be used as an input for model inversion to retrieve surface properties.

Despite the majority of atmospheric correction methods which assume a Lambertian surface for vegetation canopies, our results indicate that most vegetation canopies are non- Lambertian and this would affect the spectral radiance captured by a remote sensing instrument under different conditions, in particular solar zenith angles. Consequently, the retrieval of vegetation properties would be influenced considerably. Our findings of canopy surface reflectance anisotropy are in agreement with generally recognized non-Lambertian characteristics of natural surfaces (Gerstl and Simmer, 1986). Further, the change of the computed anisotropy index during the course of this research (Figs. 13 and 14) reveals the fact that surface anisotropy has to be taken into account not only in normal periods, when vegetation components are dominant but also during drought conditions (from mild to moderate and severe stress), when the soil background and

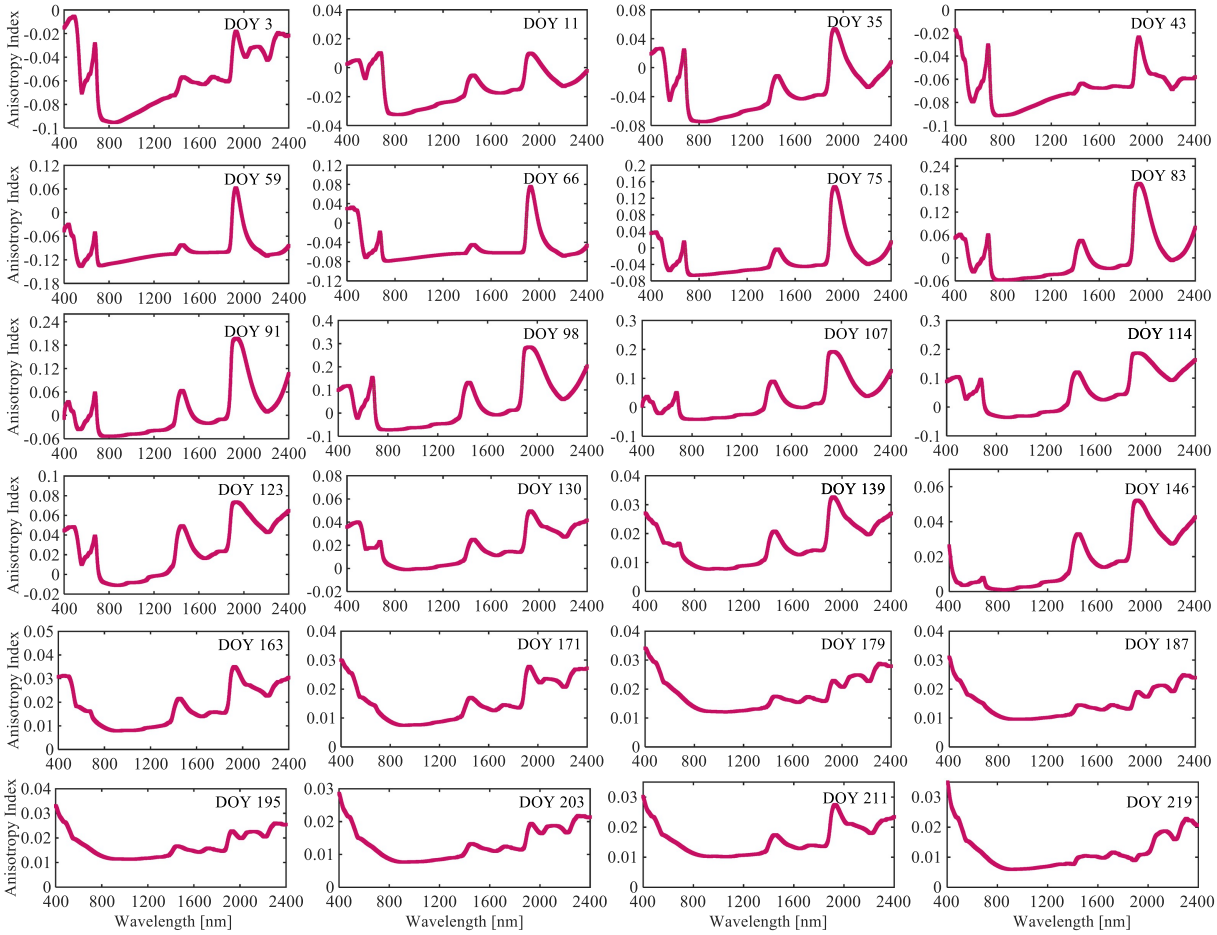


Fig. 13. Anisotropy index computed from the best simulated reflectance factors on different days at the Vaira site during the episode. It should be noted that the vertical axis range varies for different time periods.

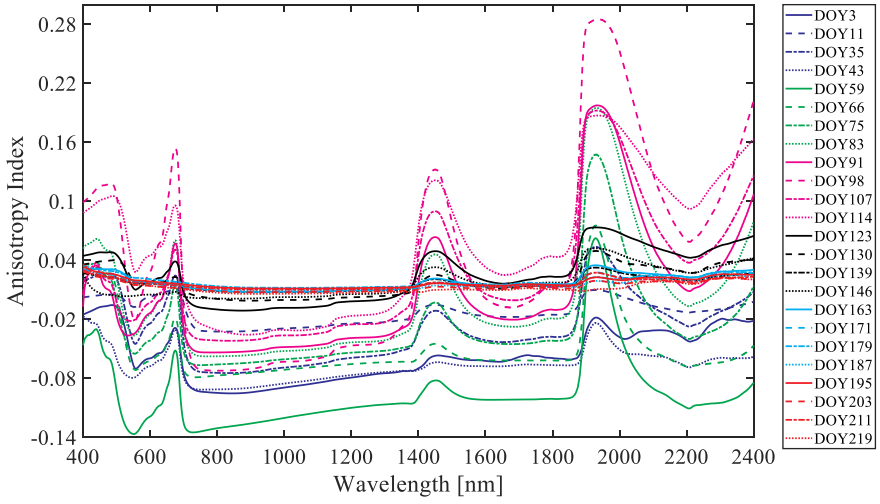


Fig. 14. Time series of computed anisotropy index and its variations during the episode at the Vaira site in 2004. It should be noted that 4 subplots (each row) of Fig. 13 are shown by an identical color but with different line styles.

plant senescence materials are dominant in the field. However, the largest effects of surface anisotropy were observed when the vegetation was in maximum growth conditions in the middle of the drought episode. Later, it decreased in concert with the decline of soil moisture content and vegetation coverage. For instance, comparing the trends in retrieved vegetation properties (Fig. 11) and the computed anisotropy index (Fig. 14), one can follow how vegetation properties are related to the canopy anisotropy changes during the episode. For instance, the maximum values of *LAI*, C_{ab}, and C_w were retrieved at the middle of the mild stress period when higher values of anisotropy index are observed, while the minimum values of *LAI*, C_{ab} and C_w were obtained later at severe stress period when lower values of the index are observed. From a modeling perspective, a canopy consisting solely of horizontal leaves is Lambertian if the soil is Lambertian as well, so the $LIDF_a$ parameter

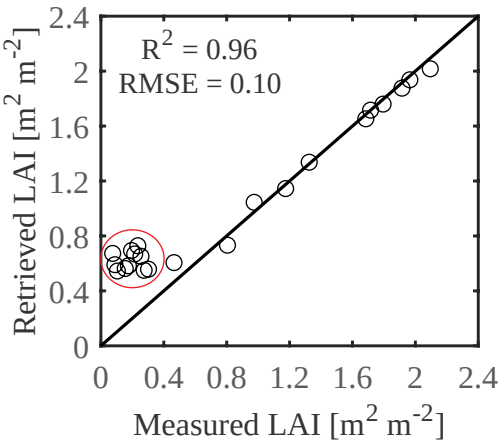


Fig. 15. The measured and retrieved LAI during the episode at the Vaira site in 2004. An ellipse is containing "too low" *LAI* values.

would generate the greatest anisotropy when it would become considerably less than unity. In Fig. 17 we observe that the retrieved $LIDF_a$ is minimum on DOY 60, at the beginning of the mild stress period when LAI, C_{ab} and C_w are still high.

Lower residuals after spectral fitting for TOA radiance ($0.006 < \text{RMSE} < 2.2\,\text{Wm}^{-2}\,\text{sr}^{-1}\,\mu\text{m}^{-1}$ for all cases) demonstrate the capability of the coupled model to reproduce the observed spectra under different environmental conditions (Fig. 11). Overall, a remarkable correspondence was observed between time series of simulated and observed TOA radiances during the episode. This can be explained by the fact that (1) we scaled the dry soil spectra by means of measured soil moisture values to give a realistic soil reflectance for different days (Fig. 12) and, therefore, the background contribution was effectively taken into account; (2) we used the a priori information (e.g., *LAI* and soil moisture) to constrain the inversion, reducing the size of parameter space and ill-posedness; (3) we considered anisotropy effects in the forward modeling instead of assuming a Lambertian surface (Figs. 13 and 14), highlighting the anisotropy influence and canopy non-Lambertian behavior. This offers indisputable and very powerful evidence for taking the anisotropic surface reflection into account in time series image analysis and RT modeling; and (4) we found the best fitting atmospheric properties during the episode for different days with different magnitudes of the atmospheric transfer functions (T-18 system) for different atmospheric conditions (Fig. 9). A wrong assumption about atmospheric properties will result in the wrong computation of TOA radiance data and, consequently, possible errors in spectral fit and the biased retrieval.

5.3. Time series of retrieved properties

Visual inspection of the retrieved properties trend during this drought episode reveals that all vegetation properties responded to drought stress, but in different ways, with two general categories. For instance, retrieved *LAI*, C_{ab} and C_w showed upward trends from normal condition up to the middle of mild stress period during the episode, while downward trends were observed for the rest of the episode (i.e., moderate and severe stress period). This pattern was opposite for C_{dm}, C_s and *LIDF*, which decreased initially in the normal condition up to the middle of mild stress, reached to minimal values in the middle and increased afterwards in moderate and severe stress period. The trend of vegetation properties observed during this episode is in agreement with a water stress laboratory experiment conducted by Bayat et al. (Bayat et al., 2016a) who demonstrated similar responses of grass to drought stress and reported considerable grass response to a drought experiment as an increase of C_{dm} and C_s, followed by a declining C_{ab}, C_w and *LAI*. Moreover, the error propagation analysis demonstrated that the possible digitization (sensor) noise embedded in the TOA radiance data does not have significant effects on the retrieved properties. In other words, this analysis indicated that the retrieval results were reliable due to the small error propagated from the assumed sensor noise.

In addition, comparing the trend over time of the retrieved vegetation properties with the trend of measured soil moisture decline during the drought episode demonstrates that there was a time delay in vegetation response (i.e., 16 days for *LAI*, 31 days for C_{ab} and 41 days for C_w) to the soil moisture deficit since the soil moisture started declining from DOY = 59 while vegetation properties decline observed from DOY 75 (*LAI*), DOY 90 (C_{ab}) and DOY 100 (C_w). Furthermore, the C_{dm} and C_s started increasing from DOY 100 (Fig. 17). This may be attributed to the fact that vegetation stress does not manifest when soil moisture deficit is below a certain limit or that water is available deeper in the root zone, but as water stress intensifies the plants will be affected.

It should be noted that the available dataset does not allow us to separate soil moisture deficit effects from those of senescence or the phenological cycle, following the natural vegetation growth cycle, since that seems not to be possible in the absence of a reliable reference (unstressed) case. Doing so would require repeating the analysis for multiple years and retrieving long-term time series of vegetation properties. In this way, one will be able to identify anomalies due to soil moisture deficit (water stress) from the "average" phenology.

5.4. Implications for multi-sensor time series synergy studies

We coupled various RT models in a bottom-up TOA approach in order to simulate time series of Landsat (TM5 and ETM7) TOA radiances and directly retrieve and map land surface properties. The results look promising and demonstrate the potential of our approach for simulation of time series of other satellite sensors data where the environment, atmospheric and canopy characteristics are varying. The coupled RT model could capture these effects from time series of the observations adequately. For instance, in our case the soil moisture deficit effects on TOA radiances and vegetation properties could be detected in a prolonged drought episode using the coupled RT models. However, there is potential to improve the resolution (spatial, temporal and spectral) and, therefore, to gain more information from the Sentinel missions using our approach. Sentinel-2 provides optical information in 13 bands (covering VIS, NIR and SWIR regions) with spatial resolutions of 10, 20 and 60 m and an average revisiting frequency of 5 days (Drusch et al., 2012). Thus, it offers a unique opportunity to use more of the information together by applying our coupled approach in an operational way to deliver high spatial resolution (of 10 and 20 m) maps of water stress effects on vegetation properties over time. This can be beneficial for both academic (e.g., remote sensing, eco-hydrology, and plant physiology communities) and application oriented (e.g., agriculture and farming) sectors to understand vegetation response to climate change.

The proposed approach may seem to be complex for operational application at first glance. However, to ensure some level of realism, one has to consider the surface – atmosphere system carefully by coupling a set of RT models to allow the retrieval of surface properties directly from TOA radiance data. The most important advantages of this approach are that (1) a separate atmospheric correction of the satellite images is not necessary, (2) the anisotropic surface reflection can be taken into account effectively and (3) the original observations of satellites (Level-1, which are freely available even for new sensors) can be utilized directly. From our point of view, these advantages can pave the way for the operational application of our approach for large multi-scene areas, albeit considering the computation time. The NO approach for the inversion of the proposed coupled model takes about 4–5s per pixel on a normal PC. This computation time can be reduced considerably using supercomputer facilities.

In addition, our findings have an important implication for multi-sensor time series synergy studies. The coupled RT model used in this study produces the atmospheric transfer functions and surface reflectance factors in the optical domain from 400 nm to 2400 nm with a spectral resolution of 0.1 or 1 cm^{-1} in wavenumber units (determined by MODTRAN), which at 1000 nm means 0.01 nm or 0.1 nm in wavelength units, respectively. This provides the user with the opportunity to resample the spectra into multiple sensors bands based on known sensor's spectral response functions. In other words, there is no need to make major changes and apply a different modeling approach in case of using various sensor data sets. The only difference is in the last step where the user needs to resample the model's outputs based on specific spectral response functions. We believe that this can be considerably beneficial for multi-sensor approaches as well as time series studies in different applications.

6. Conclusions

A forward modeling TOA radiance approach has been applied to accurately simulate an annual time series of Landsat (TM5 and ETM7) data and to retrieve vegetation properties during a drought episode in space and time. Verifying the performance of different components of the coupled model proves that BSM is able to fairly well reproduce the moist soil reflectance, RTMo is able to reproduce anisotropic TOC vegetation reflectance spectra and MODTRAN is able to accurately reproduce the observed TOA radiance spectra during this drought episode. Thus, it is concluded that the coupled use of these RT models, as a "bottom-up" approach, could be considered as a proper alternative for traditional "top-down" TOC approaches making use of atmospheric correction. The optimization approach for the inversion of the proposed coupled model takes about 4–5s per pixel on a normal PC which can be reduced considerably using supercomputer facilities.

The proposed approach is relevant for time series image analysis. Certainly in time series of satellite observations over a given target, the dependence of the target's reflectance on the solar zenith angle may interfere with the correct retrieval of true surface properties as a function of time. In addition, variations of the aerosol load in the atmosphere induce variable fractions of diffuse incident sky radiation, for which the target has its own reflectance, the hemispherical-directional reflectance factor (HDRF). This means that in time series analyses, also the atmospheric variability as a function of time has to be taken into account. Even with the best possible atmospheric correction, the

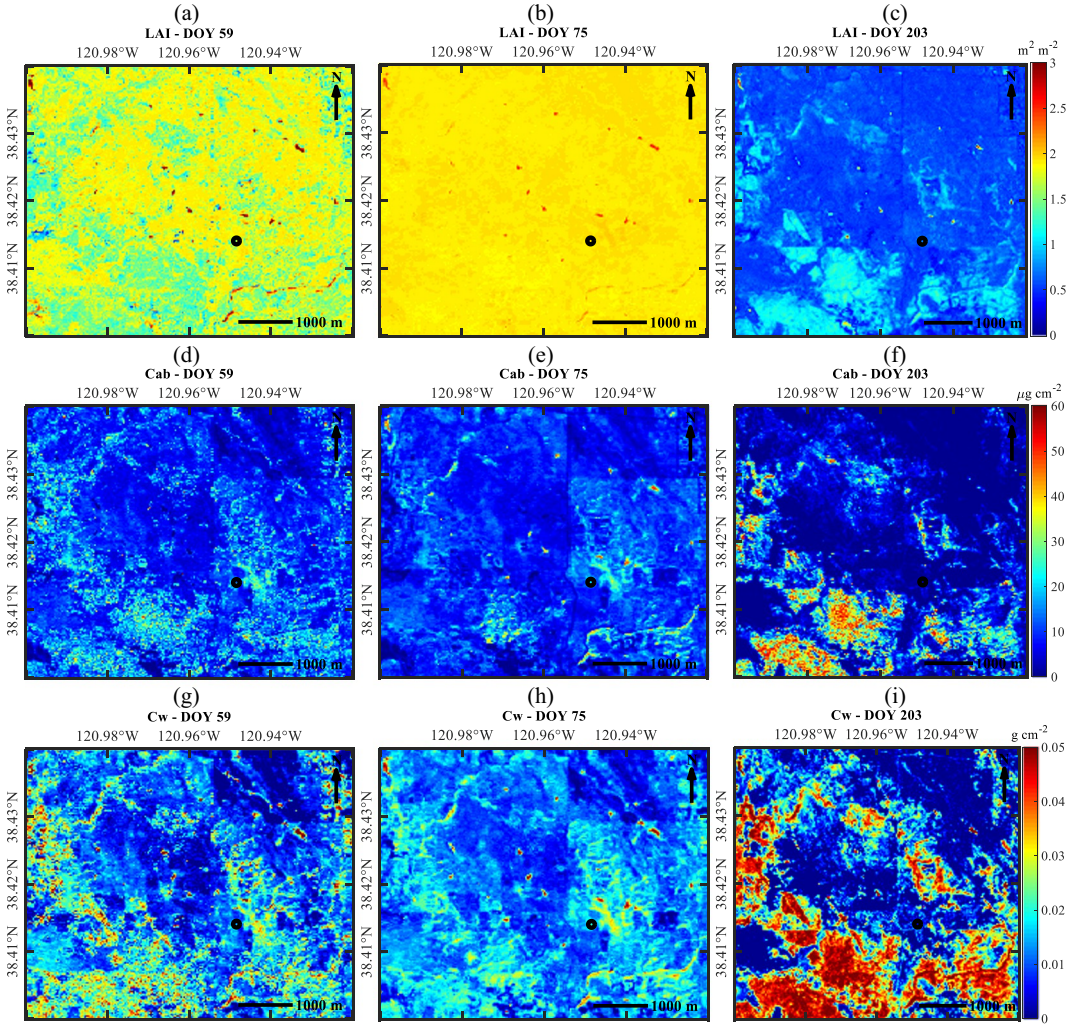


Fig. 16. Landsat maps of retrieved properties for three days during this drought episode: (a, b, c) *LAI*; (d, e, f) C_{ab}; (g, h, i) C_w; (j, k, l) C_{dm}; (m, n, o) C_s; (p, q, r) $LIDF_a$. Left panels (a, d, g, j, m, p) show the retrieved properties maps for DOY 59; middle panels (b, e, h, k, n, q) show the retrieved properties maps for DOY 75 and right panels (c, f, i, l, o, r) show the retrieved properties maps for DOY 203 in 2004. The black circle shows the location of the Vaira site. The reader is referred to Table 2 for the definitions of the surface properties.

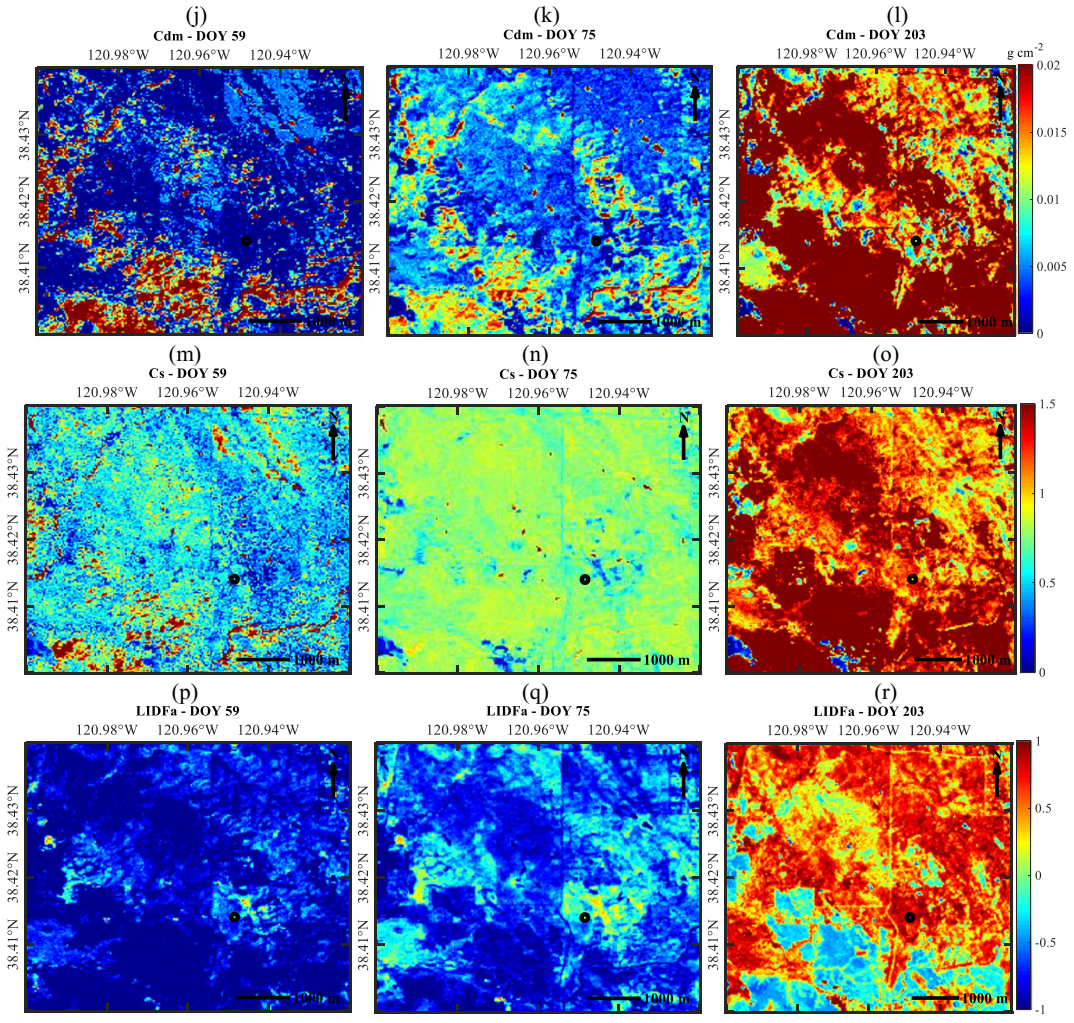


Fig. 16. (*continued*)

assumption of isotropic surface reflectance would be inconsistent with RT models that predict anisotropic reflectance behavior. The computed anisotropy index highlights the importance of taking the surface anisotropic reflection into account for simulation of TOA radiance data and provides further evidence that the vegetation canopy is non-Lambertian.

Moreover, simulated TOA radiance spectra can be sampled by the spectral response functions of arbitrary hyperspectral and multispectral sensor in space. This means that a multitude of optical sensors can give their combined inputs for time series of retrieved biophysical surface properties, thus creating a much denser temporal sampling than would be possible for separate single sensors. This is a key requirement to enable exploring time series of various satellite observations to full extent, unlocking the informative power of combined earth observation data, expanding the number of potential earth observation data users, and allowing earth observation data to become an essential asset for environmental monitoring purposes. Overall, the proposed TOA approach lowers the barrier to time series analysis of satellite images and allows users to obtain desired surface information directly and quickly to support the utilization of the growing volume of earth observation data. From an operational point of view, our TOA approach can be of great help for embedding the time series of available (and also upcoming) satellite observations in operational programs to support management objectives, enable reporting, to inform decision (policy) makers and hopefully benefit our society.

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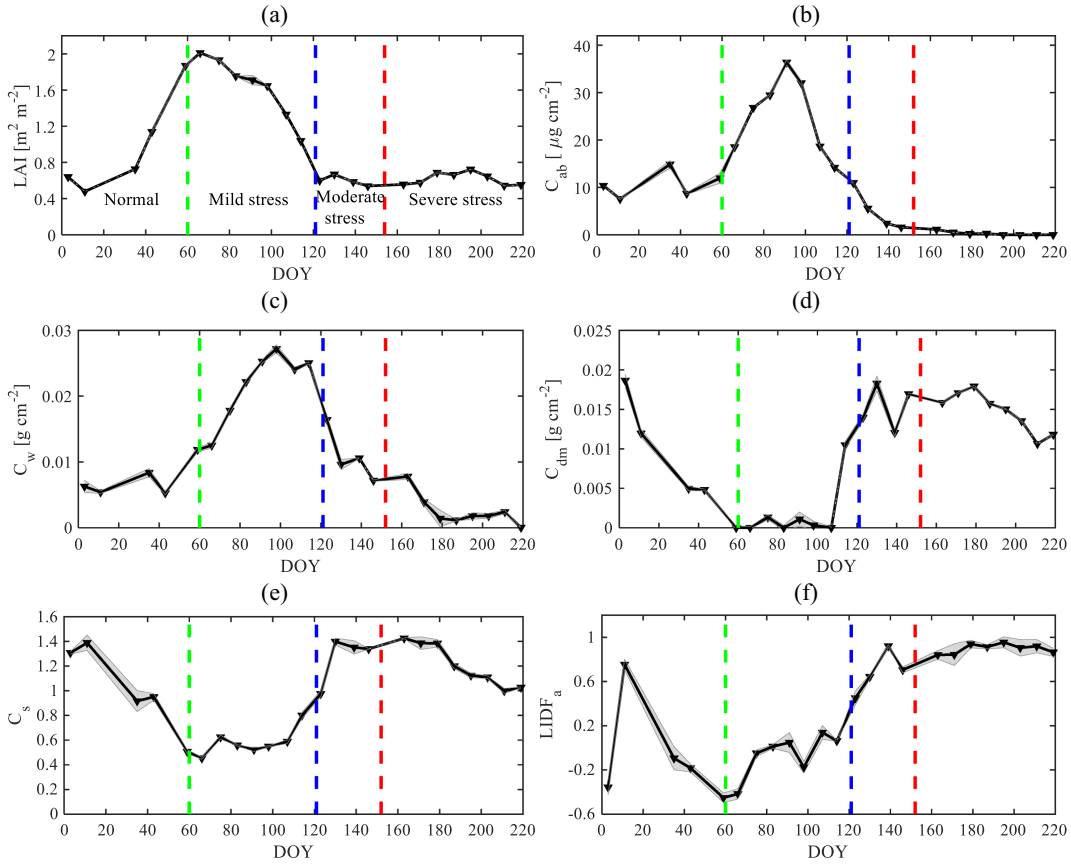


Fig. 17. Landsat retrieved properties variations at the Vaira site during the selected drought episode: (a) *LAI*; (b) C_{ab}; (c) C_w; (d) C_{dm}; (e) C_s; (f) $LIDF_a$. Error areas show the uncertainty (standard deviation) in the vegetation properties caused by the uncertainty of the Landsat-observed TOA radiance. Various drought conditions (i.e., normal, mild stress, moderate stress and severe stress) are separated by the color dotted lines.

Appendix A. Atmospheric, sensor geometric and satellite image properties

Table 1
Input parameters used in this study to describe the real atmospheric and sensor geometric conditions for MODTRAN 5 simulations.

DOY	Sensor	H_2O [g cm^{-2}]	O_3 [DU]	SZA [degree]		SAA [degree]	
				Image[a]	Vaira[b]	Image[a]	Vaira[b]
3	ETM7	0.58	0.29	65.7	65.6	155.5	154.8
11	TM5	1.7	0.29	65.9	64.9	151.9	153.4
35	ETM7	1.2	0.31	60.5	60.3	150.4	149.6
43	TM5	0.82	0.31	59.2	59.2	146.6	145.7
59	TM5	1.02	0.32	53.9	53.9	143.9	142.9
66	TM5	1.75	0.32	51.4	50.7	142.8	143.4
75	TM5	1.36	0.33	48.1	48.0	141.2	140.1
83	ETM7	1.37	0.33	43.3	43.6	142.9	141.8
91	TM5	1.37	0.34	41.9	41.9	138.3	137.5
98	TM5	2.01	0.34	39.4	58.7	136.9	137.0
107	TM5	1.19	0.34	36.7	36.9	134.9	133.3
114	TM5	0.85	0.34	34	33.4	133.1	133.6
123	TM5	1.44	0.34	31.6	31.7	130.5	128.7
130	TM5	1.8	0.34	29.9	29.3	128.4	128.8
139	TM5	1.62	0.33	28.3	28.5	125.6	123.6
146	TM5	2.02	0.33	27.4	26.6	123.4	123.7
163	TM5	1.67	0.32	25.1	25.3	122.0	120.2
171	TM5	1.35	0.32	26.4	26.7	118.4	116.5

(*continued on next page*)

Table 1 (*continued*)

DOY	Sensor	H_2O [g cm^{-2}]	O_3 [DU]	SZA [degree]		SAA [degree]	
				Image[a]	Vaira[b]	Image[a]	Vaira[b]
179	ETM7	1.96	0.31	25.6	25.8	120.5	118.7
187	TM5	2.62	0.30	27.4	27.7	118.7	116.8
195	TM5	1.31	0.30	27.2	27.3	122.1	120.4
203	TM5	2.1	0.30	29.2	29.5	121.9	120.0
211	TM5	1.6	0.30	29.59	29.7	126.4	124.7
219	TM5	1.57	0.29	31.9	32.1	127.3	125.6

[a] These values are extracted from Landsat image metadata file.

[b] These geometric variables are calculated for time of satellite overpass at the Vaira site by using NOAA Solar Calculator (*https://www.esrl.noaa.gov/gmd/grad/solcalc/*).

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